Phenotype-Aware Imitation Learning for Small, Oriented Object Detection in Remote Sensing

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***Abstract*—Imitation learning has emerged as a promising paradigm for detecting small, arbitrarily oriented objects in remote sensing imagery. However, existing methods largely focus on global feature alignment between object instances and high-quality exemplars. This often leads to overhomogenized features across instances and distortion of intrinsic semantic structures, which in turn introduces geometric ambiguities in orientation, shape, and scale. Such issues are especially detrimental in remote sensing, where small objects exhibit significant variation and complex spatial distributions. To address these limitations, we propose a phenotype-aware imitation learning (PIL) framework that explicitly preserves geometric diversity and semantic structure for accurate object localization. PIL consists of two components: a geometric disentangling branch and a semantic structure registration branch. The geometric disentangling branch learns instance-specific deformation fields that adaptively normalize orientation, scale, and shape variations between exemplars and instances. By disentangling geometric differences, this branch allows the imitation process to focus on meaningful semantic discrepancies. The semantic structure registration branch further improves alignment by decoupling imitation into registration and imitation spaces. In the registration space, instances align their semantic structures by modeling spatial relations within the exemplar's global context. In the imitation space, features that have been geometrically normalized and semantically aligned perform pixelwise imitation, supervised by a mixed-grained contrastive loss (MC-loss) that balances both local and global semantic cues. To mitigate geometric detail loss in cross-scale imitation, we also introduce a contextual micro-to-macro feature pyramid network (CM^2-FPN). This module leverages low-layer spatial detail to complement the contextual information of large objects in high-layer features. Extensive experiments on DOTA, DIOR-R, and SODA-A datasets show that PIL-Net consistently outperforms existing methods. The performance gains are especially significant in challenging settings dominated by small, densely distributed, and arbitrarily oriented targets.**

***Index Terms*—Arbitrary-oriented small-object detection, feature pyramid network (FPN), imitation learning, remote sensing images.**

I. Introduction

ARBITRARY-ORIENTED object detection in remote sensing imagery is a fundamental task in Earth observation, aiming to locate and classify rotated targets of interest [1]. It plays a critical role in various applications, including geographic information system mapping, environmental monitoring, intelligent transportation, and urban planning [2], [3], [4], [5]. Remote sensing images are typically captured from satellites, aircraft, or uncrewed aerial vehicles (UAVs) with a wide field of view, which introduces a major challenge: numerous small objects appear with arbitrary orientations and limited spatial resolution [6], [7], [8], [9], [10].

Standard feature extraction pipelines, such as multiscale convolutional backbones or feature pyramid networks (FPNs) [11], [12], are generally optimized for scale-invariant object representation. However, these pipelines inevitably compromise the representation of small objects, leading to geometric phenotype distortion and semantic identity ambiguity [13]. In contrast, large objects usually retain richer visual cues and more discriminative features [6]. This discrepancy has motivated research efforts that attempt to bridge the representational gap between large and small instances by leveraging their intrinsic correlations.

Existing approaches can be broadly categorized into two groups: super-resolution-based methods [14], [15], [16] and feature imitation frameworks [17], [18], [19], [20]. The latter has recently become the dominant direction, as it enables small instances to mimic large exemplars through global feature alignment supervised by similarity constraints, while avoiding the complex generative architectures required by the former. Despite their success, these methods often treat the exemplar and instance features as globally aligned vectors, which homogenizes intraclass representations and distorts spatial semantics. As illustrated in Fig. 1, large exemplars dominate the alignment process, suppressing fine-grained geometric cues of small instances and degrading the reliability of geometric regression. Moreover, the coarse global imitation strategy flattens each instance and exemplar into a single vector, distorting spatial semantic correspondence during imitation. Consequently, the global imitation strategy introduces geometric ambiguities that hinder precise localization of arbitrarily oriented targets [21], [22].

Received 13 February 2026; accepted 13 March 2026. Date of publication 18 March 2026; date of current version 27 March 2026. This work was supported in part by the National Natural Science Foundation of China under Grant 62271336, in part by the Key Research and Development Program of Sichuan Province under Grant 2024YFHZ0289, and in part by the Open Fund of Key Laboratory of the Ministry of Education on Artificial Intelligence in Equipment under Grant 2024-AAIE-KF04-03. *(Corresponding author: Xiaohai He.)*

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Digital Object Identifier 10.1109/TGRS.2026.3675321

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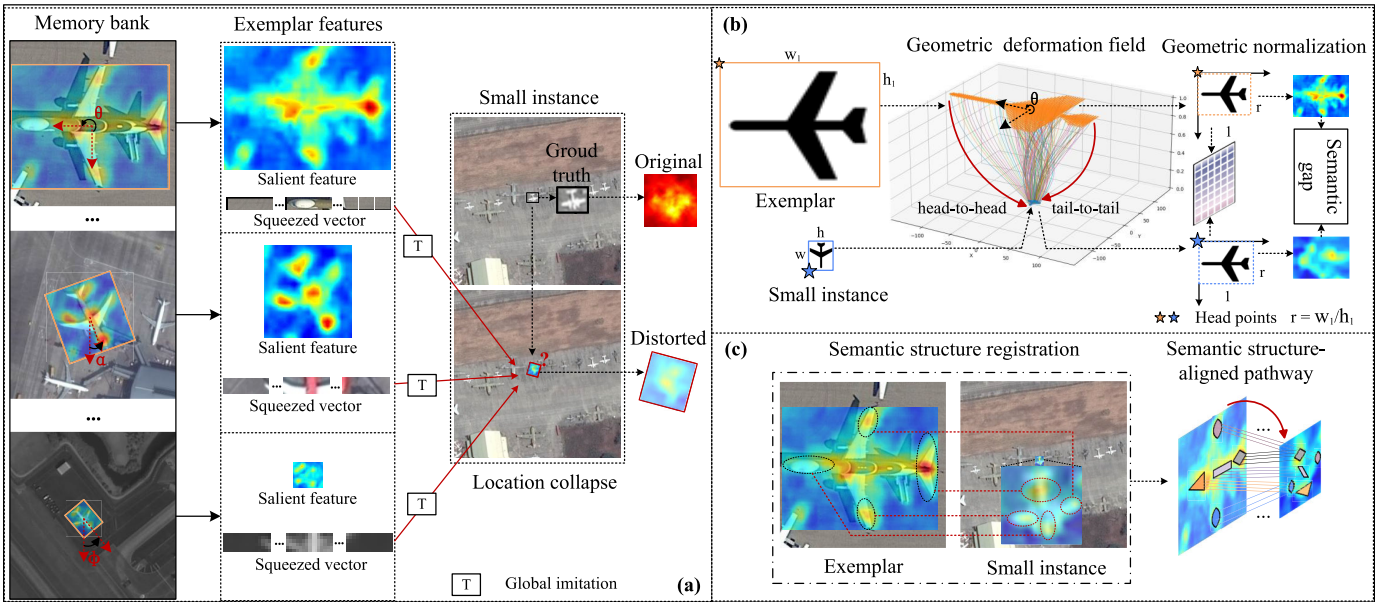


Fig. 1. Illustration of geometric ambiguities in conventional global imitation frameworks and the proposed solution. (a) Traditional global feature imitation tends to homogenize intraclass features, leading to geometric ambiguity and localization collapse. (b) Proposed geometric disentangling branch decouples geometric discrepancies from semantic gaps prior to imitation, preserving diverse geometric characteristics. (c) Semantic structure registration branch establishes a structure-aligned imitation pathway, mitigating spatial semantic distortion during feature alignment.

Cross-scale imitation further exacerbates these issues within FPN-based detectors. The abstraction gap between deep semantic-rich layers (optimized for large objects) and shallow geometric-rich layers (critical for small objects) results in geometric detail loss during feature aggregation [23]. Therefore, a mechanism capable of preserving geometric fidelity and structural diversity is essential for reliable small-object representation in cross-scale imitation.

Another important but often overlooked component in imitation learning is the exemplar memory bank. Existing frameworks typically assume a direct correlation between object scale and feature quality, selecting only large-scale exemplars to construct offline memory banks [17], [19], [20]. This assumption restricts exemplar diversity and prevents adaptive end-to-end optimization. Although [18] introduces an online quality-aware selection strategy, it depends heavily on underoptimized backbone features during early training stages, leading to unreliable exemplar selection. Thus, an adaptive and stable memory updating strategy remains necessary for robust imitation learning.

To overcome these challenges, we propose *PIL-Net*, a unified network that introduces a *phenotype-aware imitation learning (PIL)* mechanism to preserve geometric diversity and structural semantics for accurate-oriented small-object detection. Here, we borrow the term phenotype to denote an instance's observable appearance factors, including geometric characteristics (orientation, scale, and shape) and structural characteristics (intrainstance semantic structure). As shown in Fig. 1, the core PIL module consists of two branches: a geometric disentangling branch and a semantic structure registration branch, which jointly model both geometric discrepancies and intrainstance semantic structure within each instance–exemplar pair. The geometric disentangling branch constructs learnable deformation fields that normalize orientation, scale, and shape discrepancies between exemplars and instances, thereby disentangling geometric variation and allowing the imitation process to focus on semantic differences. The semantic structure registration branch decouples imitation into registration and imitation spaces. In the registration space, spatial semantic correlations are progressively established to align semantic structures between instance and exemplar pairs. In the imitation space, geometrically normalized and semantically aligned features perform pixelwise imitation guided by a mixed-grained contrastive loss (MC-loss).

Furthermore, a recurrent evolving memory bank (REM-Bank) is introduced to maintain a set of reliable exemplars that adapt dynamically to the evolving feature distribution, ensuring consistent and stable imitation supervision. To further complement PIL, we design a contextual micro-to-macro FPN (CM^2-FPN), which enhances geometric fidelity during cross-scale imitation by refining intralayer structured context (IC) under auxiliary segmentation guidance and reconstructing interlayer channels in a bottom-up manner.

The main contributions of this work are fourfold.

1) We introduce a *PIL* mechanism, composed of a geometric disentangling branch and a semantic structure registration branch. This design decouples geometric discrepancies from semantic gaps, preserving diverse spatial structures and reducing feature homogenization during imitation.
2) We develop a *REM-Bank* for stable and adaptive exemplar selection, along with an *MC-loss* to guide pixelwise imitation of semantically aligned and geometrically normalized features.
3) We propose a *CM^2-FPN* that enhances cross-scale feature imitation by refining intralayer contextual structure and reconstructing interlayer geometric details.

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4) We demonstrate superior performance on three challenging benchmarks (DOTA, DIOR-R, and SODA-A), particularly in detecting small, densely distributed, and arbitrarily oriented objects.

II. Related Work

A. AOOD in Aerial Images

Arbitrary-oriented object detection (AOOD) has been extensively studied in aerial imagery, where targets often exhibit diverse orientations, scales, and aspect ratios. To generate tightly fitted oriented bounding boxes (OBBs), AOOD methods explicitly model target orientation and can be broadly divided into two paradigms: angular parameterization and geometric representation learning.

In angular parameterization, early works directly regress rotation angles [7], [22], [24], [25], [26], [27] or corner coordinates [28], [29], [30]. However, these approaches suffer from angle periodicity and corner-order ambiguity, which impede optimization stability. Later methods, such as angle classification [31], [32], alleviate these issues by discretizing orientation spaces, while COBB [33] introduces a continuous representation to ensure smooth optimization. In contrast, geometric representation learning approaches, such as Oriented RepPoints [34], PolarDet [35], and Oriented R-CNN [36], model object orientation using adaptive points, polar coordinates, or midpoint offsets, providing greater geometric flexibility and robustness.

Although significant progress has been made, detecting small, arbitrarily oriented objects remains highly challenging. Their limited spatial extent reduces the reliability of geometric regression and weakens semantic distinctiveness. These challenges highlight the need for learning paradigms capable of transferring reliable geometric and semantic knowledge from high-quality exemplars to improve small-object representation.

B. Imitation Learning for Small-Object Detection

Small- or weak-object detection continues to be a long-standing challenge in remote sensing, mainly due to limited feature representation and low signal-to-noise ratios [6], [14], [18], [37]. In contrast, large objects typically contain richer visual cues and more distinguishable semantic structures. To reduce this disparity, imitation learning has emerged as an effective paradigm that transfers feature knowledge from large exemplars to small instances.

Early approaches, such as Kim et al. [20], enhance small pedestrian detection by concatenating their features with those of large-scale exemplars. Wu et al. [17] introduce a self-mimic learning strategy that aligns small-object features with those of large objects using an offline feature bank constructed via k-means clustering. Supervised contrastive imitation frameworks further extend this idea. CFINet [18] employs an auxiliary contrastive branch to align small RoI features with high-quality exemplars within the same category, while EGCL [19] maintains an offline pedestrian feature bank for contrastive optimization. Beyond general object detection, PCLDet [38] applies prototypical contrastive learning to fine-grained remote sensing tasks, enhancing category separability.

Despite these advances, existing imitation learning methods face three major limitations when applied to arbitrary-oriented small-object detection. First, indiscriminate global feature alignment often overhomogenizes intraclass features and distorts spatial semantics between exemplars and instances, introducing geometric ambiguity and misaligned localization. Second, most approaches rely on offline or early-stage memory banks that cannot adapt to evolving feature distributions, limiting exemplar reliability and the continuity of end-to-end training. Third, cross-scale feature imitation frequently overlooks the abstraction gap between semantic-rich deep features and geometry-rich shallow features, leading to the loss of fine-grained geometric details crucial for small-object localization. These limitations motivate the need for a more phenotype-aware imitation framework that explicitly disentangles geometric and semantic differences while maintaining stable exemplar supervision.

C. Multiscale Feature Fusion in FPNs

FPNs construct multiscale feature representations through a top-down pathway with lateral connections, allowing detectors to capture objects across different scales. However, simple additive fusion cannot effectively reconcile the semantic gap between feature levels: shallow layers contain abundant geometric details but limited semantic abstraction, while deeper layers encode high-level semantics but lose fine-grained geometry.

To address this, many FPN variants improve cross-scale information exchange and fusion. PANet [39] adds a bottom-up path, and BiFPN [40] adopts learnable weighted fusion. AugFPN [41] reduces interlevel discrepancies via consistent supervision and residual augmentation. Other designs reweight top-down connections for tiny objects [42], introduce dual-path attention-guided compensation [43], or enhance foreground-aware fusion via channel and spatial attention and adaptive aggregation [44].

For multiscale feature fusion in remote-sensing object detection, LMFF-MFFE [45] optimizes cross-scale feature propagation and designs multireceptive field refinement to balance computational cost and small-object feature representation. AMSFNet [46] introduces a multirouting scale-adaptive fusion module that aligns multistage features to a unified scale before multiplicative fusion for scale-robust remote-sensing detection. MSNet [47] enhances cross-level interaction in the pyramid through fusing high-resolution features with zoomed-out low-resolution context and injects the mixed-scale global semantics into FPN lateral connections. PSWP-DETR [48] introduces a scalewise differential module (SDM) for cross-scale fusion, where it aggregates multireceptive field features and emphasizes edge and local texture cues via edge-specific kernels. MFDAFF-Net [49] proposes a multiscale frequency-aware FPN (MF-FPN) that performs iterative cross-scale fusion using a dual-attention-guided adaptive fusion operator and further applies a wavelet-inspired frequency module (WFM) to enhance high-frequency object details. InfoFPN [5] leverages pixel shuffle operations from image super-resolution to convert channel information into spatial detail.

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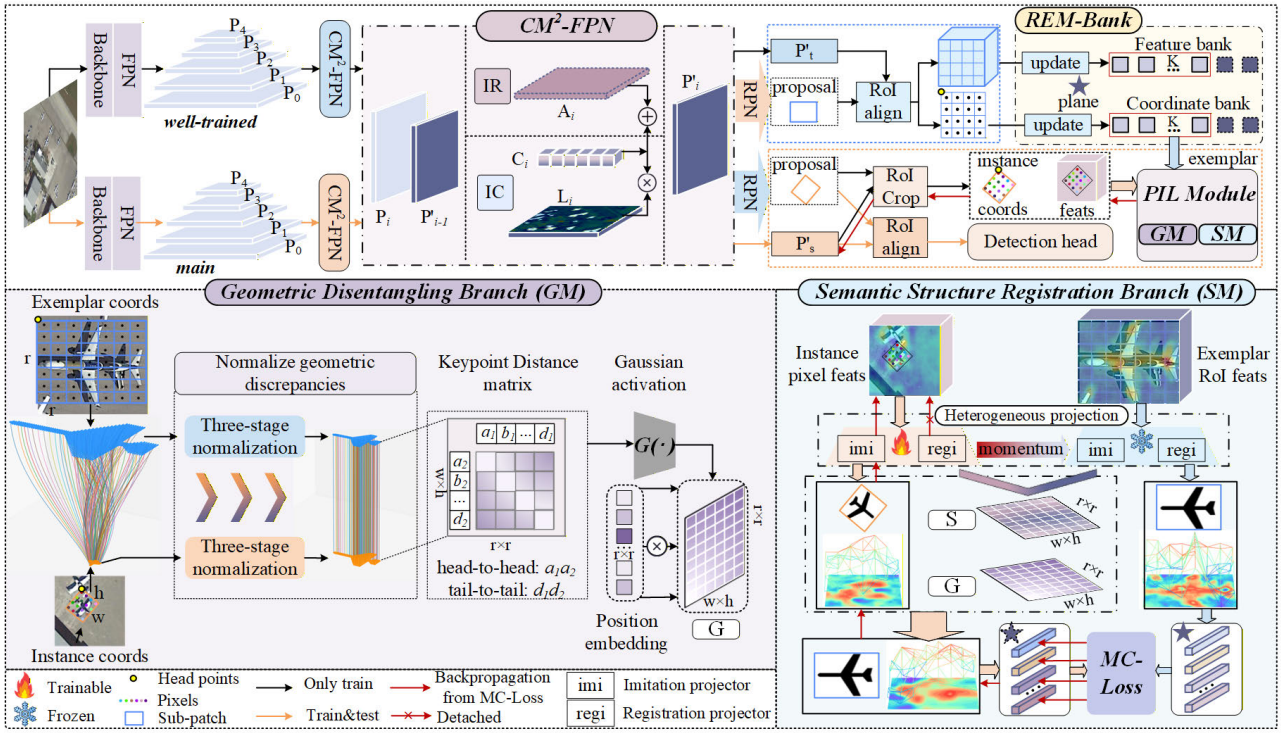


Fig. 2. Overview of PIL-Net, a rotation-aware two-stage detector enhanced with PIL. The core PIL mechanism includes a GM disentangling branch and an SM structure registration branch, which work jointly to preserve geometric diversity and align semantic structures across exemplar–instance pairs. These branches are supported by a REM-Bank and supervised by an MC-loss, which enables fine-grained, structure-aligned imitation. Additionally, the CM^2-FPN improves cross-scale feature consistency. It comprises two modules: interlayer channel reconstruction (IR) and IC refinement. Matrices S and G denote the semantic structure correspondence and learnable geometric deformation fields, respectively.

In addition, related multiscale designs also appear in other remote-sensing settings. CD-Lamba [50] targets bitemporal remote sensing change detection and builds a spatial multiresolution pyramid with a weight-shared Siamese backbone. MaDiNet [51] focuses on SAR target detection and leverages an FPN neck with deformable convolutions in middle and high pyramid layers to better handle large-scale variation and irregular target shapes.

While the above multiscale fusion strategies effectively improve cross-layer semantic alignment and multiscale feature representation through pathway redesign, attention-guided aggregation, frequency enhancement, or context enrichment, few of them focus on preserving geometric details for cross-scale feature imitation. As a result, small-object geometry can be weakened or even overridden by dominant high-level semantics and background noise. This gap motivates our CM^2-FPN, which integrates segmentation-guided intralayer purification with bottom-up interlayer reconstruction to progressively purify intralayer features and explicitly inject refined low-level geometric details into higher-level semantics.

III. Methodology

A. Overview

To address the geometric ambiguities that arise in imitation learning-based arbitrary-oriented small-object detection, we propose a unified framework, termed *PIL-Net*. As illustrated in Fig. 2, PIL-Net builds upon a two-stage arbitrary-oriented object detector and integrates PIL with enhanced multiscale feature representation.

First, the input image is fed into the backbone and a standard FPN to extract multiscale feature layers. Then, the feature layers are refined by the CM^2-FPN in the main detector, which purifies intralayer contextual structure and complements interlayer geometric details. In parallel, a well-trained teacher detector with the same feature extraction architecture generates high-quality RoI features via RoIAlign to update a REM-Bank, providing stable and distribution-consistent exemplars for imitation during training.

Based on the enhanced multiscale features from the main detector, RoIAlign is applied for detection, while RoI Crop is used to extract instance pixelwise features and pixel coordinates for imitation learning. The core PIL mechanism consists of two complementary branches. *geometric (GM) disentangling branch* operates on the instance pixel coordinates and the coordinates of exemplar RoI subpatches to learn normalized deformation fields to eliminate geometric discrepancies (e.g., scale, shape, and orientation). *Semantic (SM) structure registration branch* uses the corresponding exemplar subpatch features and instance pixel features to align semantic structures between exemplars and instances. This design prevents the homogenization of diverse geometric characteristics and promotes structure-aware imitation. Finally, the imitation process is supervised by an MC-loss, which propagates gradients to the instance pixelwise features through the disentangled and registered feature paths while suppressing instance-level noisy.

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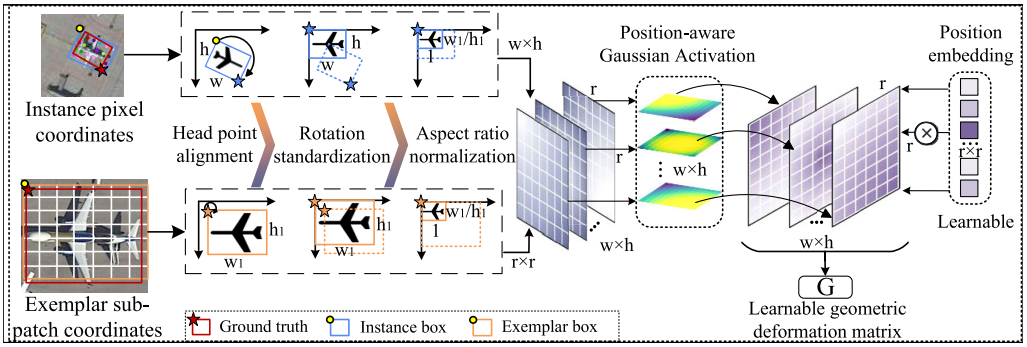


Fig. 3. Process of the geometric disentangling branch within the PIL mechanism.

B. PIL Mechanism

1) Geometric Disentangling Branch: To preserve diverse geometric characteristics, we normalize geometric phenotype discrepancies before feature imitation, disentangling instance–exemplar geometric discrepancies from semantic gaps. Consequently, the subsequent imitation focuses on transferring semantic information under geometry-aligned correspondence and reduces the risk of introducing geometric bias from shape-mismatched exemplars. To this end, we construct a learnable keypoint deformation field between each instance and exemplar, which first derives a dense keypoint correspondence matrix and then adaptively refines it through learnable position embeddings.

As illustrated in Fig. 2, we process each proposal of main detector in two parallel streams: RoIAlign features are fed to the detection head, while pixel-level features and their coordinates are cropped for fine-grained alignment and imitation. Within parallel well-trained model branch, RoI features and the centroid coordinates of each RoI subpatch are extracted via RoIAlign. These features serve to update the learned bank detailed in Section III-B4.

Building on the coordinate information, we construct the dense keypoint distance matrix where physically corresponding points (e.g., the "head" point of two ships or the "tail" point of two vehicles) are assigned the smallest distances, while noncorresponding pairs receive larger values. To achieve this, the normalization process proceeds in three stages: head point alignment, rotation standardization, and aspect ratio normalization.

As illustrated in Fig. 3, we align the physical "head" points of instance and exemplars with the "head" point of their corresponding ground truth, which is defined as the first point in the coordinate vector. For each proposal, we generate eight candidate polygons by cyclically permuting the predicted vertices as the starting point and arranging the sequence in both clockwise and counterclockwise orders. The final head-aligned polygon is selected according to the sum of Euclidean distances between each vertex of candidate polygon $v_{i,k}^{st}$ (from an instance or an exemplar) and the corresponding vertex $v_{i,k}^{g}$ of the ground truth polygon, formulated as follows:

$$d\left[v_i^{st}, v_i^{g}\right] = \sum_{k=0}^{3} \left\| v_{i,k}^{st} - v_{i,k}^{g} \right\|_2 \tag{1}$$

where i denotes the ith candidate polygon, k denotes the kth vertex, and $st \in \{s, t\}$ indicates an object from instance set (s) and exemplar set (t).

Next, we introduce instancewise rotation standardization for each instance and exemplar to ensure orientation consistency across all instance–exemplar pairs. This process eliminates redundant rotational correspondences in many-to-many imitation. Specifically, taking the first vertex (aligned head point) $\hat{v}_0^{st}$ as the origin and the long side vector $(\Delta x_{\text{long}}^{st}, \Delta y_{\text{long}}^{st})$ as rotation reference, we construct a rotation matrix RT to transform each coordinate p_i^{st} until the long side is parallel to the x-axis. This process is formulated as follows:

$$\hat{p}_i^{st} = p_i^{st} - \hat{v}_0^{st} \tag{2}$$

$$\theta^{st} = \arctan\left(\Delta y_{\text{long}}^{st} / \Delta x_{\text{long}}^{st}\right) \tag{3}$$

$$RT = \begin{bmatrix} \sin\theta^{st} & \cos\theta^{st} \\ \cos\theta^{st} & -\sin\theta^{st} \end{bmatrix} \tag{4}$$

$$\begin{bmatrix} x_i^{st\prime} \\ y_i^{st\prime} \end{bmatrix} = RT \begin{bmatrix} x_i^{st} \\ y_i^{st} \end{bmatrix} \tag{5}$$

where $p_i^{st} = (x_i^{st}, y_i^{st})$ denotes the ith pixel coordinate within the instance bounding box or centroid coordinates of the exemplar's ith RoI subpatches.

In aspect ratio normalization, each cross-scale instance–exemplar pair is projected into a unified geometric space via coordinate normalization, enabling the measurement of scale-invariant relative geometric distances. For exemplars, the transformed centroid coordinates normalized by their long-side length. For instances, the transformed pixel coordinates are first normalized, then rescaled to match the aspect ratio of the learned exemplar.

Following the three-stage alignment process, the normalized dense keypoint distance matrix ΔD of each imitation pair is computed using the instance coordinates $(x_i^{s''}, y_i^{s''})$ and exemplar coordinates $(x_j^{t''}, y_j^{t''})$ in the unified geometric space, defined as follows:

$$\Delta D_{i,j} = \begin{bmatrix} x_i'' \\ y_i'' \end{bmatrix} - \begin{bmatrix} x_j^{t''} \\ y_j^{t''} \end{bmatrix} = \begin{bmatrix} dx \\ dy \end{bmatrix}. \tag{6}$$

Here, i and j index the $w \times h$ instance points and $r \times r$ exemplar points within its RoI, respectively.

To normalize geometric discrepancies, we map the normalized dense keypoint distance matrix ΔD of each

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instance–exemplar pair into a similarity space to align geometric keypoints, as shown in Fig. 3. Given the spatial variation of pixel characteristics, boundary pixels often carry fine-grained contour information and therefore require precise local alignment, whereas central pixels correspond to more homogeneous regions that benefit from broader, smoother alignment. Accordingly, we apply a position-adaptive Gaussian activation to each normalized distance map of size $r \times r$ corresponding to each pixel of the instance, where the standard deviation σ_i is defined by the inverse of the maximum value of the ith distance matrix

$$\sigma_i = \frac{\sqrt{2}}{\max_j |\Delta D_{i,j}|_2 \times \alpha}. \tag{7}$$

Here $\sqrt{2}$ is the global maximum distance, and α is a scaling factor. Finally, boundary pixels with larger normalized distances produce smaller σ and sharper decay. In contrast, central pixels yield larger σ, enabling smoother decay. The final Gaussian similarity map of each pixel is expressed as follows:

$$Gr_i = \exp\left(-\frac{|\Delta D_i|_2^2}{2\sigma_i^2}\right). \tag{8}$$

Furthermore, since the bounding-box-based deformation (rotation, translation, and scaling) can only model coarse geometric variations, but fails to capture fine-grained object contours, we generalize the dense keypoint similarity map into a high-dimensional, learnable space. Specifically, we introduce a set of learnable position vectors $E \in \mathbb{R}^{r \times r}$, which are embedded into the matrix Gr_i

$$G_i = EGr_i. \tag{9}$$

In the subsequent imitation stage, the keypoint correspondence G_i is used to reweight RoI-cropped instance pixel features into a geometry-aligned representation for instance–exemplar imitation. However, the geometric disentangling branch does not explicitly distinguish foreground pixels from backgrounds, and may be sensitive to proposal quality when a proposal contains substantial background regions. We further introduce a semantic structure registration branch that constrains semantic structure-aligned imitation within object regions, thereby mitigating physical structure distortions and concentrating imitation on the target rather than background.

2) Semantic Structure Registration Branch: To align spatial semantic correspondence between instance and exemplar while guiding imitation on the target regions during imitation, PIL module introduces a heterogeneous projection mechanism to decouple imitation process into registration and imitation spaces.

In registration space, we establish a spatial semantic correspondence matrix, guiding pixel features of small instance to attend to exemplar regions with similar semantic structures. As shown in Fig. 4, instance pixel features F_s and exemplar subpatch features F_t are projected into parallel embedding layers (Emb_k and Emb_q) through parallel one-layer perceptions, generating key tokens (K) and query tokens (Q), respectively. The spatial semantic correspondence matrix S is then computed using these key and query embeddings

$$S = W_q^t F_t W_k^s F_s \tag{10}$$

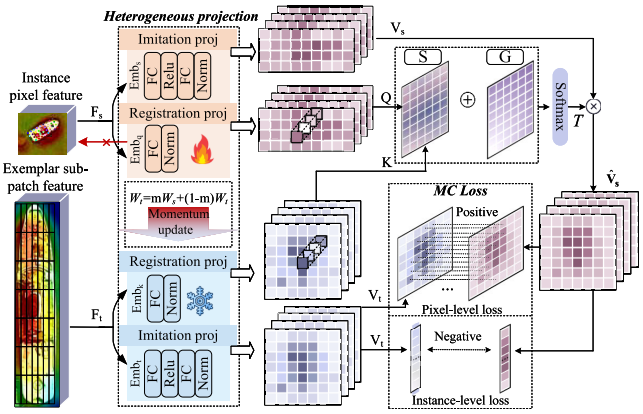


Fig. 4. Network structure of PIL mechanism's semantic structure registration branch.

where W_k^s and W_q^t denote the parameters of the fully connected (FC) layers for key and query projections, respectively. The correspondence matrix S measures the spatial semantic similarity of each sub-patch in the exemplar and each pixel in the instance, where larger values indicate stronger semantic matches. Since background regions usually lack object-specific matchable structures, they tend to yield low correspondence values in S, thereby being largely excluded from the imitation.

To ensure training stability, we freeze gradient propagation of the exemplar branch, preventing interference with the parameter updates in main detection. Specifically, only the key encoder parameters are updated via backpropagation. The query encoder parameters are updated through a momentum-based rule

$$W_q^t = mW_k^s + (1-m)\, W_q^t \tag{11}$$

where $m \in [0, 1)$ is a momentum coefficient (set to 0.999 following [52]).

In imitation space, we leverage the spatial semantic correspondence matrix (S) and geometric deformation matrix (G) to reconstruct instance feature. As illustrated in Fig. 4, instance pixel features and exemplar subpatch features are projected into the unified imitation space via parallel two-layer perceptions (Emb_s and Emb_t), where the parameters of Emb_t are also updated via momentum update rule [52] based on Emb_s for stable training. Meanwhile, these two matrices are fused into a hybrid alignment factor T by elementwise addition, followed by a Softmax activation layer.

And then, the hybrid alignment factor T operates as a weight matrix to prealign the instance embeddings. Notably, to ensure that the prealignment process only optimizes the information propagation pathway for imitation learning, the network responsible for generating T is decoupled from the detection backbone's gradient flow via gradient detachment. This design enables high-quality semantic transfer while preserving geometric characteristics.

3) Mixed-Grained Contrastive Loss: Based on these pre-registered imitation pairs, we propose an MC-loss to supervise fine-grained feature propagation between positive pairs and suppressing instance-level negative noise. Given a positive pair $(\hat{V}^s_{(i,k)}, V^t_k)$, $\hat{V}^s_{(i,k)}$ denotes the ith reconstructed instance feature

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guided by the kth exemplar V_k^t. We formulate a pixel-level similarity function to minimize the distances across all r^2 aligned pixel pairs, thereby ensuring geometric-normalized and semantic structure-aligned imitation. In contrast, for each negative sample B_i, where pixel-level correspondence is unnecessary, we maximize coarse instance-level similarity to suppress noise interference. The similarity metrics are defined as follows:

$$M_{(i,k)}^{st}\left(\hat{V}_{(i,k)}^{s}, V_k^t\right) = \frac{1}{\tau}\sum_{p=1}^{r^2}\left\langle \frac{\hat{V}_{[(i,k),p]}^{s}}{\left\|\hat{V}_{[(i,k),p]}^{s}\right\|_2}, \frac{V_{[k,p]}^{t}}{\left\|V_{[k,p]}^{t}\right\|_2}\right\rangle \tag{12}$$

$$M_{(i,k)}^{sB}\left(\hat{V}_{(i,k)}^{s}, B_j\right) = \frac{1}{\tau}\left\langle \frac{\hat{V}_{(i,k)}^{s}}{\left\|\hat{V}_{(i,k)}^{s}\right\|}, \frac{B_j}{\left\|B_j\right\|}\right\rangle \tag{13}$$

where τ denotes the temperature hyperparameter, set to 0.07 following [52], and p is the pixel index.

We further refine the sampling strategy for the contrastive loss. Prior works [18], [38] define positive samples as the RoI features from the same category, and negative samples as the RoI features from different categories. However, since RoI features are extracted from first-stage detector feature maps, complex cross-category negative samples conflict with the first-stage binary classification task, potentially introducing irrelevant semantic noise. Therefore, we exclusively use K background exemplars as negative samples.

Finally, the contrastive loss for each student instance is computed as the sum of cross-entropy losses over all positive and negative pairs (14). Specifically, for the ith student instance, we construct a label vector $[1_K\|0_K]$ as $\mathcal{Y}_i$, where K exemplars from the same category are labeled as positive and K background exemplars as negative. Simultaneously, we concatenate the positive and negative similarity scores (M_i^{st} and M_i^{sB}) into a prediction vector M_i. Additionally, each student instance's category score is used as the loss weight W_i to emphasize more confident proposals, mitigating the impact of noisy instances. The overall contrastive loss is the sum of losses over all of student instances

$$L_{IL}^{i} = -W_i\left[\sum_{pk=1}^{K}\mathcal{Y}_{(i,pk)}\log M_{(i,pk)} + \sum_{nk=K+1}^{2K}\left(1-\mathcal{Y}_{(i,nk)}\right)\log\left(1-M_{(i,nk)}\right)\right] \tag{14}$$

where pk and nk represent the positive and negative pair indices, respectively.

4) Recurrent Evolving Memory Bank: To ensure stable and effective imitation learning, it is crucial that the exemplar bank not only provides high-quality exemplars but also remains aligned with the current data distribution to prevent learning collapse caused by large distribution gaps. To this end, we design a REM-Bank for object detection. Given the instability of proposal quality in early training, we leverage an independent, well-trained teacher network to extract high-quality exemplars from each mini-batch and recurrently update the historical memory. These exemplars are then used for the subsequent imitation stage, where they are projected into the imitation space by a momentum-updated projector to stabilize the supervision signal, as described in Section III-B2. Here, the "well-trained teacher" is defined as the final checkpoint trained with the same training schedule and optimization hyperparameters as the baseline detector, and it is kept frozen during the main training.

Algorithm 1 Construction of REM-Bank

Input: S: Sets of positive exemplars (boxes, scores, features, coordinates, aspect ratios, gt);
S_{back}: Background RoI features;
C_N: Number of categories;
K: Queue length of feature banks;
Output: $Q_f^{\text{pos}}, Q_c^{\text{pos}}, Q_r^{\text{pos}}$: Category-wise positive feature, coordinate, and aspect ratio queues;
Q_f^{neg}: Negative feature queue;

Feature Bank Initialization:
for $cls \in \{1, 2, \ldots, C_N\}$ **do**
  $Q_f^{\text{pos}}[cls] \leftarrow \text{Normalize}(\text{Randn}(K, c, r, r), \text{dim} = 1)$
  $Q_c^{\text{pos}}[cls] \leftarrow \text{Normalize}(\text{Randn}(K, r^2, 2), \text{dim} = 1)$
  $Q_r^{\text{pos}}[cls] \leftarrow \text{Normalize}(\text{Randn}(K, 2), \text{dim} = 1)$
end for
$Q_f^{\text{neg}} \leftarrow \emptyset$
Positive Bank Update via Hybrid Sampling:
for $cls \in \{1, 2, \ldots, C_N\}$ **do**
  $\mathcal{I}_{cls} \leftarrow [i \mid i, s \in \text{enumerate}(S), s.\text{gt.cls} = cls]$
  if $\mathcal{I}_{cls} \neq \emptyset$ **then**
    $R_{cls} \leftarrow S[\mathcal{I}_{cls}].\text{score}[cls]$
    $feat_{cls} \leftarrow S[\mathcal{I}_{cls}].\text{feat}$
    $coor_{cls} \leftarrow S[\mathcal{I}_{cls}].\text{coor}$
    $ratio_{cls} \leftarrow S[\mathcal{I}_{cls}].\text{ratio}$
    $\mathcal{I}_s \leftarrow \text{Sort}(R_{cls}, \text{ascending})$
    $Q_f^{\text{pos}}[cls] \leftarrow \text{Concate}(Q_f^{\text{pos}}[cls], feat_{cls}[\mathcal{I}_s])[-K]$
    $Q_c^{\text{pos}}[cls] \leftarrow \text{Concate}(Q_c^{\text{pos}}[cls], coor_{cls}[\mathcal{I}_s])[-K]$
    $Q_r^{\text{pos}}[cls] \leftarrow \text{Concate}(Q_r^{\text{pos}}[cls], ratio_{cls}[\mathcal{I}_s])[-K]$
  end if
end for
Negative Bank Update:
$Q_f^{\text{neg}} \leftarrow \text{RandSample}(S_{\text{back}}, K)$

As outlined in Algorithm 1, since our PIL requires a memory bank that supports geometric-aware imitation, REM-Bank jointly maintains RoI feature, coordinate, and aspect ratio banks, enabling synchronized updates of semantic and geometric information. These banks are first initialized by constructing category-specific queues using normal random distributions. Specifically, feature queues Q_f are initialized as CN random tensors with dimensions (K, c, r, r), where CN denotes the number of categories, K is the queue length, and (c, r, r) represent the size of each RoI feature, respectively. Centroid coordinate queues Q_c and aspect ratio queues Q_r are initialized as two sets of CN random tensors with dimensions $(K, r \times r, 2)$ and $(K, 2)$, respectively. These queue tensors pass through channelwise normalization to ensure consistent magnitude scales, stabilizing the training process.

In parallel with training, the well-trained teacher network generates high-quality proposals as initial learned positive exemplars (S) and negative exemplars (S_{back}). Among them,

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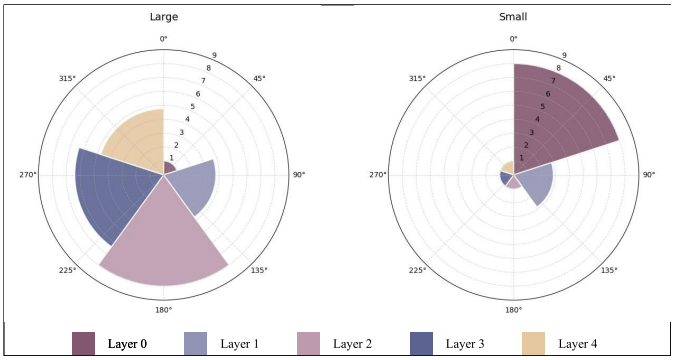


Fig. 5. Statistics on the percentage of small and large proposals at different abstraction layers. Layer 0 is the shallowest layer.

background samples are typically scene-dependent and exhibit large cross-scene variability, whereas category-consistent foreground proposals tend to be more semantic-consistent. Accordingly, REM-Bank adopts distinct update strategies for foreground and background exemplars. Foreground exemplars belonging to the same category are first aggregated and sorted by the predicted category score to ensure that the most reliable exemplars are leveraged to update the category-specific queues. These enable the accumulation of rich, semantically consistent foreground exemplars across scenes for robust imitation. Unlike positive samples, K negative samples are randomly sampled from the background proposals in current mini-batch to avoid introducing contextually irrelevant negatives from past scenes.

Finally, sorted current exemplars are enqueued into their corresponding category-specific queues, with historical exemplars replaced in a first-in-first-out manner. This quality-aware recurrent evolving mechanism ensures the feature bank remains synchronized with the training process, maintaining both high exemplar reliability and consistency with the current data distribution.

C. Contextual Micro-to-Macro FPN

As a classic architecture for multiscale detection, the FPN leverages hierarchical features to detect objects across scales: large objects are typically detected using deeper features enriched with high-level semantic information, while small objects rely on shallow features preserving fine-grained details (Fig. 5). Recent multiscale fusion approaches in remote sensing object detection primarily focus on semantic consistency and scale robustness, while rarely being designed for geometry-aware cross-scale feature imitation. Consequently, naive cross-scale feature imitation is prone to further weaken or overwhelm geometric details of small instances, under dominant high-level semantics and background noise. To address this issue, we propose a CM^2-FPN, which combines semantic template-guided intralayer purification with bottom-up interlayer channelwise reconstruction to preserve small-object geometric details during cross-scale feature imitation. It consists of an IC aggregation module and an interlayer semantic-sensitive reconstruction (IR).

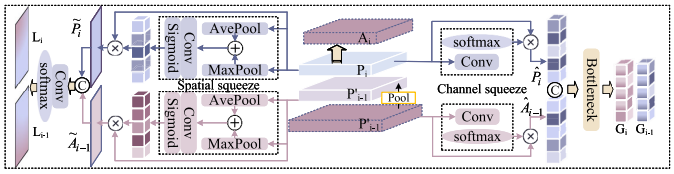


Fig. 6. Network structure of interlayer semantics-sensitive channel reconstruction (IR).

1) Interlayer Semantic-Sensitive Channel Reconstruction Module (IR): Since semantic abstraction corresponds to semantic intensity distributed along the channels [53], we model interlayer channelwise dependencies at both global and local spatial levels as interlayer context compensation vectors and reconstruction weights respectively, thereby propagating and complementing geometric details in a bottom-up way.

Given the current FPN feature P_i and the reconstructed feature at a lower level P'_{i-1}, as shown in the right part of Fig. 6, we first obtain their adaptive spatial attention weights by applying a channel squeeze operation consisting of a 1×1 convolution followed by a Softmax layer. Under the guidance of these weights, features of all spatial positions in both P_i and P'_{i-1} are aggregated as global spatial features $\hat{P}_i$ and $\hat{A}_{i-1}$. These global spatial features retain fine channel information, are then concatenated along channel dimension, and pass through a bottleneck transformation to capture long-range channelwise interaction as interlayer context compensation vectors $G_{i-1,i}$

$$\hat{P}_i = \sigma\left(W_S^P P_i\right) P_i \tag{15}$$

$$\hat{A}_{i-1} = \sigma\left(W_S^A P'_{i-1}\right) P'_{i-1} \tag{16}$$

$$G_{(i-1,i)} = W_2 \mathrm{ReLU}\left(LN\left(W_1\left(\hat{P}_i \| \hat{A}_{i-1}\right)\right)\right) \tag{17}$$

where W_S^A and W_S^P are learnable weights for generating spatial attention weights, and σ denotes the Softmax function. Equation (17) represents the bottleneck, where W_1 and W_2 are weight parameters of 1×1 convolution layers, and $\|$ denotes channelwise concatenation (consistently used throughout this article).

In parallel, as shown in the left part of Fig. 6, we capture channel attention weights through a spatial squeeze operation, which is implemented by aggregating the outputs of global average and max pooling, followed by a 1×1 convolution and a Sigmoid activation layer. Then, we broadcast the channel attention weights and computing pixelwise weighted sums to capture local spatial features with global channel information. These local spatial features are then concatenated along channel dimension and passed through a 1×1 convolutional layer to model channel interaction as interlayer reconstruction weights

$$\widetilde{P}_i = \delta\left(W_C^P\left(GAP\left(P_i\right) + GMP\left(P_i\right)\right)\right) P_i \tag{18}$$

$$\widetilde{A}_{i-1} = \delta\left(W_C^A\left(GAP\left(P'_{i-1}\right) + GMP\left(P'_{i-1}\right)\right)\right) P'_{i-1} \tag{19}$$

$$L_{(i-1,i)} = \mathrm{Softmax}\left(W_3\left(\widetilde{P}_i \middle\| \widetilde{A}_{i-1}\right)\right) \tag{20}$$

where W_C^P and W_C^A are learnable weights for modeling channel dependencies, W_3 is the weight parameter for 1×1 convolutional layer, and δ denotes the Sigmoid function.

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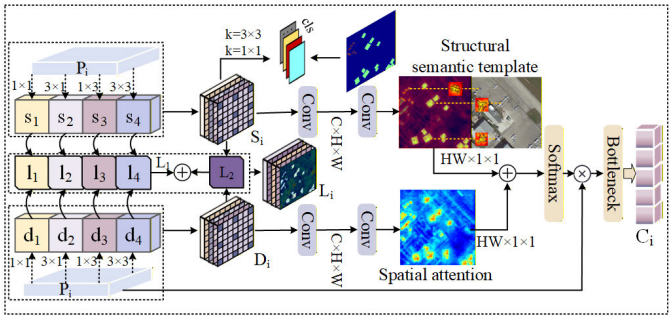


Fig. 7. Network structure of IC aggregation.

Finally, the interlayer semantic-sensitive channel reconstruction process is formulated as follows:

$$A_i = L_{i-1} \cdot \left(P'_{i-1} + G_{i-1}\right) + L_i \cdot (P_i + G_i). \tag{21}$$

2) IC Aggregation: To ensure reliable micro-to-macro feature propagation, we design an IC refinement module. Unlike detection features that are often noisy and category-mixed, segmentation inherently enforces category purity and preserves spatial structures consistent with instance boundaries [24], [53]. Motivated by this, we leverage auxiliary segmentation features to construct a context category template, which serves to purify and aggregate detection context through local and global interaction (GI) between detection and auxiliary segmentation branches.

As illustrated in Fig. 7, given the ith FPN layer is P_i, multireceptive field features are extracted and concatenated as S_i and D_i. These features are then utilized to derive multireception field local interaction and GI. In the GI, S_i is squeezed along the channel dimensions through hybrid convolution layers F_{conv} to produce a compact structured semantic template, while D_i is projected into spatial attention features. Through additive fusion of S_i and D_i, we explicitly encode structural semantic cues into the detection branch. The fusion results serve as guidance for global context aggregation, yielding the structured context feature C_i. This process is formulated as follows:

$$C_i = Bottle\left(C_i \sigma\left(F_{conv}(S_i) + F_{conv}(D_i)\right)\right) \tag{22}$$

where σ denotes the Sigmoid function, and *Bottle* denotes a bottleneck, which is applied to further enhance interchannel dependencies within this aggregated context, which uses the same structure as in Section III-C1.

In parallel, the local interaction (LI) models the pixel-level correlations between detection and segmentation features aiming to enhance the category discriminability of foreground regions and suppress semantic confusion, thereby purifying the detection branch. Specifically, based on S_i and D_i in multireception fields, we obtain a multireception field interaction feature (L_2) through a simple hybrid convolution structure. Meanwhile, since the features with the same reception have better information correspondence, we combine (s_1, s_2, s_3, s_4) and (d_1, d_2, d_3, d_4) with the same reception field to obtain the single-reception interaction features, which are fused to generate another multireception field relations (L_1). L_1 and L_2 are fused as the purified detection feature L_i. This process is formulated as follows:

$$L_1 = F\left(\left(f\left(\bigcup_{i=1}^{4} s_i\right)\right), f\left(\bigcup_{i=1}^{4} d_i\right)\right) \tag{23}$$

$$L_2 = F\left(\bigcup_{i=1}^{4} f\left(\bigcup (s_i, d_i)\right)\right) \tag{24}$$

where f denotes a 3×3 convolutional layer, $\bigcup$ denotes channelwise concatenation, and F represents the hybrid convolutional module including a 3×3 convolutional layer followed by a batch normalize and a ReLU activation, and a 1×1 convolutional layer followed by a batch normalize and a *ReLU* activation.

Finally, we integrate the resulting A_i from IR, as well as C_i and L_i from IC to reconstruct the FPN layer P_i. Specifically, as shown in Fig. 2, the GI feature C_i is combined with the interlayer channel reconstruction feature A_i, embedding semantically structured global context into the FPN. Simultaneously, the LI feature L_i is integrated through a dot multiplication operation to further purify category-discriminative representations. The reconstruction module progressively enhances intralayer semantic reliability and complements higher-layer representations with the refined lower-layer details, therefore complementing the loss of geometric details during cross-scale feature imitation.

D. MultitTask Learning Loss Function

The proposed PIL-Net incorporates multiple tasks: coarse-oriented proposals learning in RPN, refined regression and classification of the proposals, auxiliary semantic segmentation, and feature imitation. Thus, we present a comprehensive loss, which is defined as follows:

$$L = \lambda_1 L_{RPN} + \lambda_2 L_{R-CNN} + \lambda_3 L_{IL} + \lambda_4 L_{SEG}. \tag{25}$$

Here, L_{RPN} and L_{R-CNN} follow the definitions in Oriented R-CNN, each comprising classification and regression losses: cross-entropy loss for classification and smooth $L1$ loss for regression. L_{IL} denotes the contrastive imitation learning loss defined in Section III-B3, and L_{SEG} is the loss of segmentation branch. The hyperparameters λ_1, λ_2, λ_3, and λ_4 balance the contributions of individual tasks, with default values set to 1.

Notably, the segmentation branch is supervised by box-wise segmentation maps, where the foreground regions are assigned the category indices of corresponding objects (i.e., $(0, 1, \ldots, CN-1)$) and background regions assigned CN (the total number of categories). L_{SEG} is a cross-entropy loss, which is formulated as follows:

$$L^l_{SEG} = -\sum_{l \in [0:4]} \frac{1}{H_l \times W_l} \sum_{d \in F_l} \sum_{c=1}^{C} y^d_c log\left(p^d_c\right) \tag{26}$$

$$L_{SEG} = \sum_{l=0}^{4} L^l_{SEG} \tag{27}$$

where $p^d_c \in [0, 1]$ is the predicted probability of class c at location d in feature layer F_l; y^d_c is a one-hot vector, where the cth channel is 1 and all others are 0; and H_l and W_l denote the height and width of F_l, respectively.

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E. Training and Interference Protocol

PIL mechanism is used during training to perform exemplar-guided feature imitation on positive RoIs, which ensures that imitation is performed on positive proposals with sufficient target overlap. The imitation losses are backpropagated through the shared backbone to improve detection performance. During inference, we disable PIL mechanism and use the detector alone for prediction, consisting of the backbone and FPN followed by the proposed CM^2-FPN and the detection head.

IV. Experiments

In this section, we perform comprehensive experiments on oriented object detection datasets in the remote sensing field, including two mainstream benchmarks (DOTA [1] and DIOR-R [4]) and one large-scale small-object dataset (SODA-A [6]). First, we provide an overview of these datasets and evaluation metrics. Second, extensive quantitative and qualitative results on the three public datasets are presented and compared with other state-of-the-art methods to validate the performance of the proposed approach. Finally, ablation studies are conducted to assess the contributions of our PIL-Net's key components: the PIL mechanism and a CM^2-FPN.

A. Datasets and Protocols

Although our approach is primarily designed to address the challenges of small-object detection, PIL-Net does not perform scale-specific selective enhancement. Instead, it uniformly applies imitation learning to proposals across all scales, aiming to improve feature representation in a scale-agnostic manner.

Therefore, to comprehensively validate the general effectiveness of our method under various object scales, we conduct extensive experiments on two widely used multiscale datasets (DOTA [1] and DIOR-R [4]) designed for oriented object detection in aerial images. In addition, to specifically highlight its advantages in handling small objects, we further evaluate it on a dedicated arbitrary-oriented small-object benchmark (SODA-A [6]).

Following the official protocols, we use the standard train/validation/test splits. Specifically, DOTA images are cropped into 1024 $\times$ 1024 patches with 200-pixel overlap, while SODA-A images are cropped into 800 $\times$ 800 patches with a stride of 650 and then resized to 1200 $\times$ 1200.

B. Implementation Details

Experiments were performed on a single server with two NVIDIA GeForce RTX3090Ti. The stochastic gradient descent (SGD) optimizer is employed for training with a momentum of 0.9 and a batch size of 2. The proposed model is trained for 12 epochs. Weight decay was set to 0.0001 for DOTA and DIOR-R, and to 1/3 for SODA-A. The initial learning rate was 0.005 for DOTA and DIOR-R, and 0.01 for SODA-A; it was decayed by 0.1 at epoch 8 and epoch 11. In addition, we pretrain the teacher detector offline on each dataset using the same schedule and optimization hyperparameters, and employ its final checkpoint as the teacher. The teacher shares the same detector architecture as the main network and is kept frozen during the main training process, providing confidence scores and RoI features for online REM-Bank updates at each mini-batch.

C. Evaluation Metrics

To quantitatively evaluate the performance of different detectors, we adopt the widely used mean average precision (mAP) metric, derived from precision–recall analysis. In particular, for evaluation of small-object detection, we additionally employ area-based AP subsets as defined in SODA-A: AP_s, AP_{eS}, AP_{rS}, AP_{gS}, and AP_N. Specifically, these metrics correspond to AP values for instances within specific area ranges: $(0, 1024]$, $(0, 144]$, $(144, 400]$, $(400, 1024]$, and $(1024, 2000]$, respectively [6].

D. Comparison With the State-of-the-Art Methods

1) Results on DOTA: To evaluate the performance of the proposed PIL-Net, we conduct comprehensive comparisons with state-of-the-art approaches on the single-scale DOTA dataset. As shown in Table I, building upon Oriented R-CNN with a ResNet-50-FPN backbone, PIL-Net achieves mAP of 78.36%, outperforming previous studies focused on geometric–semantic modeling paradigms for oriented object detection [21], [54], [55], [56], [57]. This result demonstrates that our novel imitation-based learning framework effectively guides instances to imitate reliable and fine semantic information while preserving critical geometric robustness. Through constraining semantic structure-aligned and geometric-normalized pixel-pair feature imitation, PIL-Net establishes new SOTA performance.

To further illustrate the superior performance, we provide qualitative comparisons against three representative remote-sensing oriented detection baselines, including ARC [56] as a geometry-aware oriented detector, CFINet [18] as a coarse-to-fine approach tailored for small-object detection, and Oriented RepPoints [34] as a widely used point-based oriented detector. As illustrated in Fig. 8, we highlight challenging regions in the detection results with orange boxes, and mark the corresponding areas in the feature map visualizations with black borders. ARC tends to suffer from missed detections and orientation inaccuracies under densely arranged and low-contrast targets; Oriented RepPoints exhibits both missed detections and category confusion in cluttered backgrounds; and CFINet still fails to detect extremely low-contrast objects and may produce regression bias on small targets. In contrast, our approach yields more reliable detection with tighter localization for small oriented objects, especially in dense and cluttered remote-sensing scenes.

In addition, the feature map visualizations further reveal the response patterns underlying these results. Among them, the feature maps of comparison methods often display diffuse activations, where foreground responses are less salient and object boundaries are relatively blurred. This consequently degrade classification and regression accuracy. In contrast, our method generates feature maps that are more focused on the foreground, with a spatial distribution that aligns better with the target shape, while effectively suppressing background responses.

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TABLE I
COMPARISON WITH STATE-OF-THE-ART METHODS ON DOTA DATASET. NOTE: COLOR CODING INDICATES RELATIVE PERFORMANCE—BLUE: BEST, GREEN: SECOND-BEST

Methods	Backbone	PL	BD	BR	GTF	SV	LV	SH	TC	BC	ST	SBF	RA	HA	SP	HC	mAP
One-stage																	
RetinaNet-O[59]	R-50-FPN	88.67	77.62	41.81	58.17	74.58	71.64	79.11	90.29	82.18	74.32	54.75	60.60	62.57	69.67	60.64	68.43
R^3Det[25]	R-152-FPN	89.49	81.17	50.53	66.10	70.92	78.66	78.21	90.81	85.26	84.23	61.81	63.77	68.16	69.83	67.17	73.74
S^2A-Net[22]	R-50-FPN	89.11	82.84	48.37	71.11	78.11	78.39	87.25	90.83	84.90	85.64	60.36	62.60	65.26	69.13	57.94	74.12
DFDet[60]	R-50-FPN	88.92	79.25	48.40	70.00	80.22	78.85	87.21	90.90	83.13	83.98	60.07	66.49	68.27	76.78	58.11	74.71
FRED[57]	ReR50	89.37	82.12	50.84	73.89	77.58	77.38	87.51	90.82	86.30	84.25	62.54	65.10	72.65	69.55	63.41	75.56
RO2-DETR[61]	R-50	88.98	82.13	54.58	76.28	79.28	77.92	87.94	90.90	87.19	85.65	62.20	62.63	74.62	72.42	59.22	77.57
Two-stage																	
Scrdet++[24]	R-101-FPN	89.77	83.90	56.30	73.98	72.60	75.63	82.82	90.76	87.89	86.14	65.24	63.17	76.05	68.06	70.24	76.20
RoI-Trans.[27]	R-101-FPN	88.65	82.60	52.53	70.87	77.93	76.67	86.87	90.71	83.83	82.51	53.95	67.61	74.67	68.75	61.03	74.61
ReDet[54]	ReR-50-ReFPN	88.79	82.64	53.97	74.00	78.13	84.06	88.04	90.89	87.78	85.75	61.76	60.39	75.96	68.07	63.59	76.25
Oriented RepPoints[34]	R-101-FPN	89.53	84.07	59.86	71.76	79.95	80.03	87.33	90.84	87.54	85.23	59.15	66.37	75.23	73.75	57.23	76.52
ARC[56]	ARC-R50	89.40	82.48	55.33	73.88	79.37	84.05	88.06	90.90	86.44	84.83	63.63	70.32	74.29	71.91	65.43	77.35
UFDet[62]	R-50-FPN	89.40	83.63	53.03	72.81	78.98	81.44	87.91	90.90	87.00	84.40	64.21	69.10	70.87	71.33	63.61	76.57
ReDiffDet[63]	R-50-FPN	85.45	77.84	50.39	71.63	80.90	84.85	88.77	90.88	87.00	86.08	63.30	63.43	76.01	76.03	60.16	76.18
FPNFormer[21]	FPNFormer	87.87	81.57	59.71	78.08	79.49	85.55	88.40	90.48	83.23	86.29	67.39	69.39	77.86	76.71	62.43	78.30
ORSR[55]	ReR-50-ReFPN	89.53	85.04	57.20	75.05	77.59	83.90	87.80	90.78	88.87	86.74	63.65	66.30	78.05	70.81	69.07	78.03
Oriented R-CNN[36]	R-50-FPN	89.46	82.12	54.78	70.86	78.93	83.00	88.20	90.90	87.50	84.68	63.97	67.69	74.94	68.84	52.28	75.87
Oriented R-CNN[36]	LSKNet-S-FPN[58]	89.66	85.52	57.72	75.70	74.95	78.69	88.24	90.88	86.79	86.38	66.92	63.77	77.77	74.47	64.82	77.49
Ours																	
PIL-Net	R-50-FPN	89.43	83.63	58.38	72.83	80.46	83.31	88.16	90.89	84.23	86.60	63.65	72.00	77.01	77.69	67.16	78.36
PIL-Net	LSKNet-S-FPN	89.35	83.40	59.68	78.33	82.38	85.36	88.21	90.86	85.51	86.36	62.91	67.90	78.16	80.95	75.52	79.66

TABLE II
DETECTION ACCURACY ON DIOR-R TEST SET

Methods	**Faster RCNN-O**[64]	**Gliding Vertex**[29]	**RoI-Trans.**[27]	**MGNet**[65]	**Oriented RepPoints**[34]
mAP	59.54	60.06	63.87	66.3	66.71
Methods	**ORSR**[55]	**MutDet**[66]	**PKINet**[67]	**Oriented R-CNN**[36]	**PIL-Net**
mAP	65.18	66.8	67.03	64.63	**67.31**

These observations are consistent with our design: CM^2-FPN suppresses intralayer clutter while complementing cross-scale geometric details, while PIL enhances foreground saliency and preserves geometric characteristics and semantic structure via PIL.

To further validate the model's capability in detecting small and weak objects, we incorporate LSKNet [58], which is a lightweight backbone specifically designed for small- and weak-object detection. Results in Table I show that PIL-Net further enhance LSKNet-based Oriented R-CNN baseline on small objects, such as BR, SV, LV and SP, which increased by 1.96%, 6.68%, 6.67% and 6.48%, respectively. These gains confirm that PIL-Net synergistically leverages LSKNet's strengths in small-object detection, further elevating performance in this critical task.

Meanwhile, Fig. 9 presents qualitative detection results on the DOTA test set. These indicate that PIL-Net can achieve robust performance for weak and small instances. The abbreviations for categories are defined as follows (abbreviation-full name): PL-Plane, BD-Baseball diamond, BR-Bridge, GTF-Ground field track, SV-Small vehicle, LV-Large vehicle, SH-Ship, TC-Tennis court, BC-Basketball court, ST-Storage tank, SBF-Soccer-ball field, RA-Roundabout, HA-Harbor, SP-Swimming pool, and HC-Helicopter (the same below).

2) Results on DIOR-R: Compared with state-of-the-art methods on the DIOR-R dataset, PIL-Net achieves the highest mAP of 67.31%, as shown in Table II. All experiments use a ResNet-50-FPN backbone, with the exception of PKINet, which employs its custom PKINet-S backbone.

Meanwhile, as presented in Fig. 10, PIL-Net exhibits strong robustness in recognizing objects with high intraclass diversity, such as airports, dams, and baseball fields. Furthermore, compared with the baseline (Fig. 11), PIL-Net generates more precise and tightly fitted bounding boxes, demonstrating its

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TABLE III
COMPARISON WITH STATE-OF-THE-ART DETECTION APPROACHES ON THE SODA-A TEST SET

Paradigm	Methods	AP	AP_{50}	AP_{75}	AP_{eS}	AP_{rS}	AP_{gS}	AP_N
One-stage	S^2A-Net[22]	28.3	69.6	13.1	10.2	22.8	35.8	29.5
	DHRec[68]	30.1	68.8	19.8	10.6	24.6	40.3	34.6
	DCFL[69]	33.6	70.8	26.4	11.7	28.3	44.8	38.0
	FCOS-O[70]	32.8	70.4	26.2	10.9	27.9	43.9	38.1
	SemDiff[13]	34.3	71.4	28.3	12.3	29.0	44.5	**38.9**
Two-stage	RoI-Trans.[27]	36.0	73.0	30.1	13.5	30.3	46.1	39.5
	STD[71]	30.3	67.4	21.2	12.4	26.6	36.7	30.7
	Oriented RepPoints[34]	26.3	58.8	19.0	9.4	22.6	32.4	28.5
	DODet[72]	31.6	68.1	23.4	11.3	26.3	41.0	33.5
	CFINet[18]	34.4	73.1	26.1	13.5	29.3	44.0	35.9
	Oriented R-CNN[36]	34.4	70.7	28.6	12.5	28.6	44.5	36.7
	PIL-Net(Ours)	**36.6**	**73.7**	**31.4**	**14.6**	**31.3**	**46.9**	**38.9**

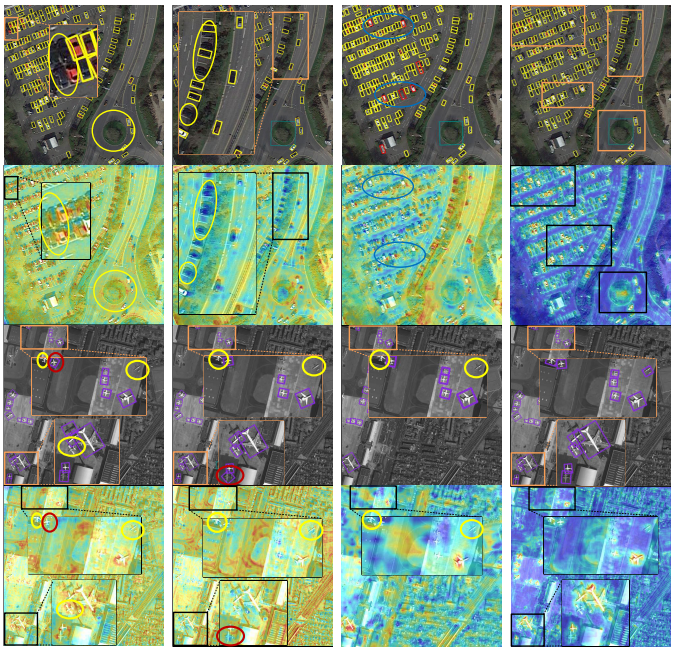


Fig. 8. Qualitative comparisons of detection results and prediction feature maps. Yellow ellipses denote missed detections, blue ellipses denote classification errors, and red ellipses denote inaccurate localization. (a) ARC. (b) CFINet. (c) O-RepPoints. (d) PIL-Net.

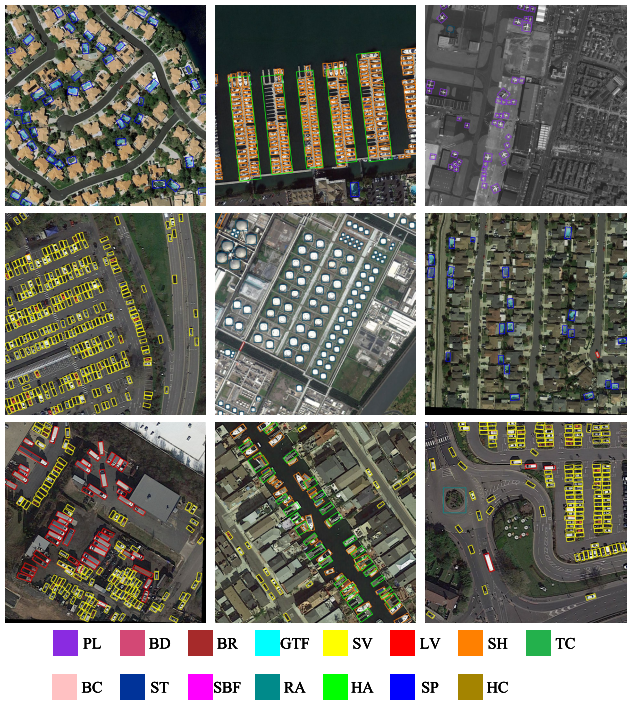


Fig. 9. Detection samples of PIL-Net on DOTA test set.

ability to better preserve geometric diversity. Overall, by incorporating PIL into the detection framework, PIL-Net captures more discriminative semantic and localization features.

3) Results on SODA-A: To highlight PIL-Net's capability in detecting size-limited objects, we employ SODA-A, a dataset specifically designed for small-object detection tasks, and conduct a comprehensive comparison with state-of-the-art representative approaches. As shown in Table III, our PIL-Net achieves state-of-the-art performance with an overall AP of 36.6%, outperforming the baseline by 2.2%. It also shows clear advantages in specific small-object detection metrics: AP_{eS} (+2.1%), AP_{rS} (+2.7%) and AP_{gS} (+2.4%).

Furthermore, PIL-Net significantly outperforms CFINet [18], a specialized small-object detection method via imitation learning, with improvements of 2.2%AP, 1.1%AP_{eS}, 1.6%AP_{rS}, and 2.8%AP_{gS}.

Qualitative detection results of the baseline method and PIL-Net on the SODA-A dataset are presented in Fig. 12, which intuitively demonstrates our effectiveness in multi-category small-object detection. Category abbreviations are defined as follows (abbreviation-full name): AP-airplane, HC-helicopter, SV-small vehicle, LV-large vehicle, SH-ship, CT-container, ST-storage-tank, SP-swimming-pool, and WM-windmill.

E. Complexity and Speed Comparison

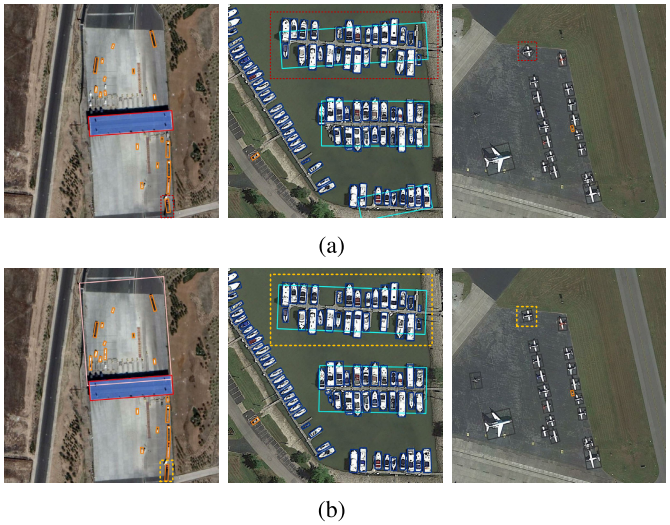
We analyze the inference-time computational cost of different methods during inference in terms of parameters, GFLOPs, and inference speed (FPS). All detectors are trained on the DOTA training set and evaluated on its validation set. FPS

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(a) (b) (c)

Fig. 10. Detection samples with high intraclass diversity of PIL-Net on the DIOR-R test set. (a) Airport. (b) Dam. (c) Baseball field.



(a)

(b)

Fig. 11. Qualitative comparison of localization accuracy on DIOR-R test set. Red and yellow rectangles are added to highlight the corresponding samples for clearer comparison. (a) Detection samples of baseline. (b) Detection samples of PIL-Net.

is measured under single-scale inference with input size 1024×1024 and batch size 1 on a single NVIDIA RTX 3090 Ti GPU. As shown in Fig. 13, incorporating PIL brings clear performance improvements while maintaining almost the same Params, GFLOPs, and FPS as the baseline (Oriented R-CNN). This is because PIL is only introduced during training and is not involved in inference, introducing negligible overhead. The increase in model complexity mainly comes from the CM^2-FPN module. Specifically, CM^2-FPN performs IC aggregation at each FPN level and conducts interlayer semantic-sensitive channel reconstruction between adjacent pyramid levels. These operations introduce additional convolutional computations and cross-level feature interactions, resulting in a significant increase in the GFLOPs and a corresponding decrease in inference speed. Overall, the proposed PIL-Net achieves higher detection accuracy with an expected trade-off in computational cost introduced by CM^2-FPN.

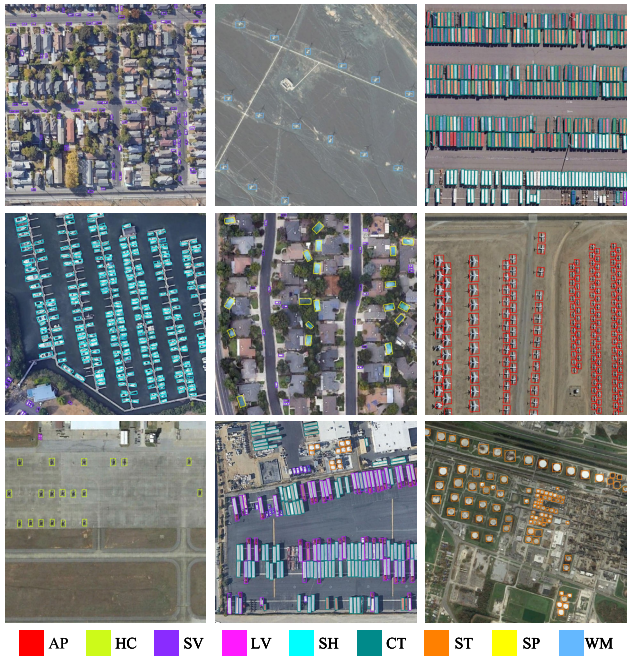


Fig. 12. Detection samples of PIL-Net on SODA-A test set.

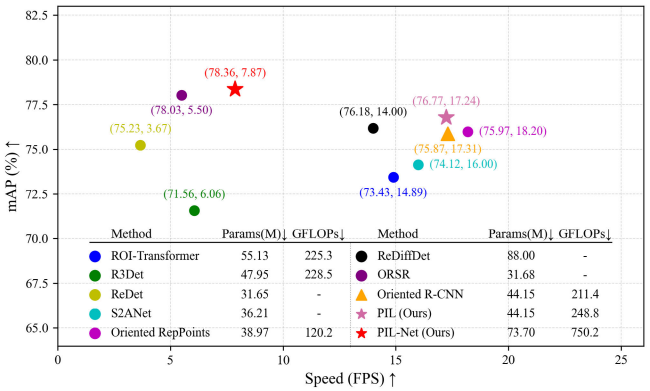


Fig. 13. Performance (mAP) versus speed (FPS), parameters, and GFLOPs comparison correspond to the inference stage on the DOTA validation set. We evaluate RoI-Transformer [27], R^{3}Det [25], S^{2}A-Net [22], Oriented RepPoints [34], ReDiffDet [63], and ORSR [55] with ResNet-50. The ReDet [54] uses ReResNet-50 as its backbone. Our baseline Oriented R-CNN [36], our PIL module, and PIL-Net also use ResNet-50 as the backbones in this comparison.

F. Ablation Study

To verify the effectiveness of each component and assess its applicability in generic detection scenarios, comprehensive ablation studies are performed on the DOTA dataset. We employ Oriented R-CNN as our baseline model due to its concise network structure and high accuracy.

1) Evaluation of Different Components: We conduct a componentwise ablation study to systematically evaluate the contribution of each module in PIL-Net. As shown in Table IV, incorporating PIL module alone yields mAP of 76.77%, demonstrating that its feature imitation mechanism effectively improves semantic discrimination and location precision. Introducing CM^2-FPN alone raises the mAP to 77.70%, highlighting its ability to refine intralayer context

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TABLE IV
ABLATION STUDY OF THE COMPONENTS IN PIL-NET ON DOTA TEST SET. ALL REPORTED RESULTS ARE PERFORMED ON THE SINGLE-SCALE DOTA DATASET

PIL	CM^2-FPN	PL	BD	BR	GTF	SV	LV	SH	TC	BC	ST	SBF	RA	HA	SP	HC	mAP
×	×	89.46	82.12	54.78	70.86	78.93	83.00	88.20	**90.90**	**87.50**	84.68	63.97	67.69	74.94	68.84	52.28	75.87
✓	×	89.41	83.26	54.54	72.93	79.17	83.53	88.28	**90.90**	85.80	85.06	**64.21**		74.69	69.66	60.78	76.77
×	✓	**89.49**	**83.79**	55.40	**77.16**	80.30	**84.97**	**88.44**	90.89	87.17	85.99	62.69	66.44	75.63	72.12	64.99	77.70
✓	✓	89.43	83.63	**58.38**	72.83	**80.46**	83.31	88.16	90.89	84.23	**86.60**	63.65	**72.00**	**77.01**	**77.69**	**67.16**	**78.36**

TABLE V
ABLATION STUDY OF SUBMODULES IN PIL MECHANISM

Components	Different Variants				
SI	✓				
GM(no embed)		✓			
GM			✓		✓
SM				✓	✓
mAP	76.29	76.45	76.47	76.62	**76.77**

and enrich high-level representations with fine-grained lower-layer geometric details. Additionally, combining CM^2-FPN and PIL module increases the mAP to 78.36%, establishing 2.49% improvement over the baseline. This demonstrates that CM^2-FPN effectively complements PIL module through compensating for geometric detail loss in cross-scale feature imitation. Moreover, our PIL-Net improves the performance of the baseline Oriented R-CNN in most of the categories, especially for BR, SV, LV, SP, and HC which contain numerous small instances.

2) Evaluation of PIL Mechanism: In the following, we present experimental validations focusing on the core components of PIL mechanism. Specifically, we conduct ablation studies to verify the effectiveness of its submodules: GM disentangling module and SM structure registration module; validate the superiority of its MC-loss; and assess the reliability of its REM-Bank, thereby systematically verifying the effectiveness of PIL mechanism.

a) Ablation study of submodules in PIL mechanism: PIL mechanism operates GM to disentangle geometric discrepancies prior to imitation learning and performs SM to align spatial semantic correspondence between instance and exemplar during imitation. To verify the effectiveness of these submodules, we conduct an ablation experiment and visualize IoU–Score distributions.

We first establish a simple imitation (SI) branch without prealignment as the baseline. As shown in Table V, introducing SI improves the baseline performance, as it guides the instance feature aligning with high-quality exemplar representations, thereby enhancing semantic discriminability. Building upon this, incorporating GM without learnable position embedding (no embed) yields 0.16% mAP improvements, supporting our hypothesis that unconditional imitation tends to geometric ambiguities, while our geometric discrepancy disentanglement effectively helps alleviate the homogenization of diverse geometric features. Combining learnable position vector with GM further improves the mAP to 76.47% demonstrating that our learnable position embedding enables an adaptive and fine-grained geometric prealignment, which is more powerful for handling and decoupling complex imitation-pair geometric variations from semantic gaps. Adding SM alone achieves 0.33% gains, indicating that semantic structure-aligned feature imitation can mitigate physical structure distortions during imitation. The combination of GM and SM achieves the optimal performance of 76.77% mAP, indicating that the PIL mechanism propagates high-quality exemplar features for proposals while preserving diverse geometric phenotypes.

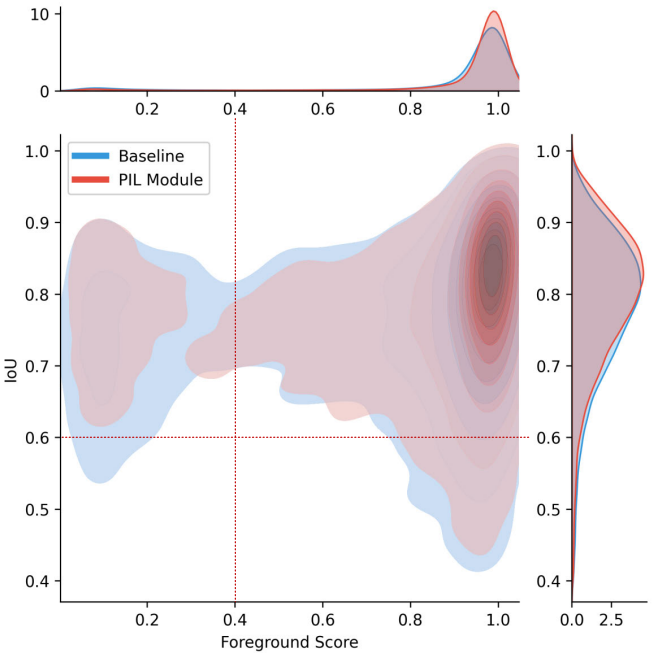


Fig. 14. Comparison of score-IoU distributions (PIL mechanism versus baseline).

To visually demonstrate the performance improvement of the PIL module, we present comparative visualizations of 1-D marginal distributions of IoU and foreground scores, as well as 2-D IoU-Score joint distributions in Fig. 14.

The 1-D IoU distribution (right marginal plot) reveals a denser concentration of proposals with IoU >0.8 compared to the baseline, indicating that PIL module generates more proposals with stronger regression potential. This improvement is primarily attributed to PIL's ability to preserve instance geometric characteristics and intrinsic semantic structures for precise localization. For foreground scores (up marginal plot), PIL mechanism exhibits a sharper, more concentrated peak around score 1.0 compared to the baseline, confirming that PIL effectively enhances discriminative feature learning by

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TABLE VI
PERFORMANCE UNDER VARYING PROPOSAL QUALITY IN DOTA VALIDATION SET

GM	SM	mAP@IoU[0.1, 0.3)	mAP@IoU[0.3, 0.5)	mAP@IoU[0.5, 1.0)
×	×	74.0	66.27	57.04
✓	×	71.41 (-2.59)	67.13 (+0.86)	58.75 (+1.71)
✓	✓	74.01 (+0.01)	68.06 (+1.79)	61.31 (+4.27)

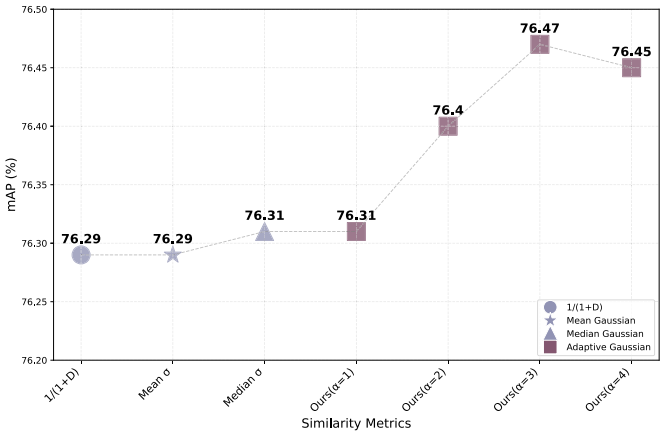


Fig. 15. Comparison of different similarity metrics.

precisely guiding each student pixel to imitate the high-quality exemplar regions through an optimized pathway.

The 2-D joint kernel density estimation plots demonstrate that PIL mechanism significantly reduces proposal density in low-confidence (<0.4) and low-IoU (<0.6) regions relative to the baseline. These discrepancies further validate that PIL's feature imitation strategy suppresses low-quality proposal generation by jointly ensuring geometric precision and improving semantic reliability.

b) Ablation study of PIL submodules under varying proposal quality: In remote sensing object detection, small targets are often surrounded by cluttered backgrounds. Proposals with localization bias may not tightly enclose the target, and thus include background pixels. To verify the robustness of PIL under such imperfect proposals, we conduct an ablation study on the DOTA validation set by grouping each ground truth instance according to the maximum IoU among its matched proposals into three bins: $[0.1, 0.3)$, $[0.3, 0.5)$, and $[0.5, 1.0)$, and report mAP within each bin. As shown in Table VI, GM improves performance in the higher-IoU bins, achieving +1.14 mAP in $[0.3, 0.5)$ and +1.71 mAP in $[0.5, 1.0)$. However, it leads to a drop of 2.59 mAP in the lowest-IoU bin $[0.1, 0.3)$. These suggest that GM relies on RoI quality and becomes less robust for background-dominated RoIs. Adding SM eliminates the low-IoU degradation observed in GM and yields consistent improvements across all bins, demonstrating that spatial semantic-aligned imitation effectively mitigates background interference in the imitation process and stabilizes feature imitation under imperfect proposals.

c) Evaluation of position-aware gaussian activation in PIL mechanism: To validate the effectiveness of our adaptive Gaussian activation, we conduct extensive ablation studies against other similarity metrics. Specifically, we compared it with a naive inverse-distance baseline $1/(1 + D)$, as well as Gaussian variants with different definitions of σ, including nonadaptive σ set to the mean or median distance, and our adaptive distance with ($\alpha \in (1, 2, 3, 4)$). As shown in Fig. 15, our adaptive Gaussian with $\alpha = 3$ achieves the best performance.

This demonstrates that, properly configured Gaussian activation significantly outperforms the simple inverse-distance formulation by providing sharper attenuation of geometrically irrelevant keypoints. Furthermore, using position-adaptive σ is superior than nonadaptive σ, indicating that localized alignment is more appropriate for fine-grained contour regions, while broader alignment benefits homogeneous central regions, and this position-adaptive property is effectively achieved by our position-adaptive Gaussian formulation, which ensures sharper decay near boundaries and smoother responses in interior areas. Simultaneously, choosing $\alpha = 3$ achieves the best trade-off in alignment region coverage.

TABLE VII
COMPARISON BETWEEN DIFFERENT IMITATION LOSSES

Module	baseline	CR-loss	CN-loss	IN-loss	MC-loss
mAP	75.89	76.29	75.98	76.24	**76.77**

d) Evaluation of MC-loss: To further validate the effectiveness of our MC-loss, we compare it with several supervision strategies. Specifically, MC-loss is designed to minimize the similarity distance between aligned pixel pairs of positive imitation pairs, while suppressing instance-level negative exemplars sampled from background proposals.

To examine the impact of feature grain, we first replace MC-loss with a coarse RoI contrastive loss (CR-loss), where coarse RoI features are directly used as contrastive positive and negative samples. As shown in Table VII, MC-loss outperforms CR-loss (76.77% versus 76.29%), demonstrating that, when integrated with PIL mechanism, MC-loss guides pixel features of instances to capture finer representations in an optimized imitation pathway, in contrast to coarse contrastive loss, which indiscriminately imitates global positive pairs and thus leads to feature excessive homogenization.

We further analyze the effect of negative sample composition. Specifically, we follow prior works and introduce a category-negative loss (CN-loss). For each instance, CN-loss samples K exemplars from different categories (including background) as negative exemplars. MC-loss surpasses CN-loss by 0.79% mAP, indicating that focusing contrastive negative samples on background features avoids irrelevant semantic noise and stabilizes the foundational representation learning in the first detection stage.

Finally, we compare MC-loss with the standard instance-level InfoNCE loss (IN-loss) [52], [73]. Our approach achieves higher performance (76.77% versus 76.24%), highlighting its superior ability to improve feature discriminability.

e) Evaluation of REM-Bank: To verify the effectiveness of our REM-Bank, we conduct ablation studies. First, we compare REM-Bank with a MoCo-style memory variant [52],

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TABLE VIII
COMPARISON BETWEEN DIFFERENT MEMORY BANKS

Module	baseline	MM-Bank	BF-Bank	REM-Bank
mAP	75.89	76.06	76.62	**76.77**

MM-Bank, where the memory bank is updated using proposal features extracted from the network being optimized. As shown in Table VIII, REM-Bank achieves an mAP of 76.77, outperforming MM-Bank (76.06 mAP), demonstrating the benefit of teacher-guided, quality-aware memory updates for the reliable memory construction in detection. Second, we analyze the foreground–background update strategies, by introducing a variant (BF-Bank) that also maintains background proposals in a memory queue, as with the foreground queues. BF-Bank yields inferior performance at 76.62 mAP, indicating that sampling background from the current mini-batch is crucial to avoid introducing contextually irrelevant negatives that hinder effective imitation learning.

f) Evaluation of imitation granularity: To evaluate the effect of imitation granularity, we evaluate proposal-level imitation (ProIL) following CFINet [18], and our pixel-level imitation (PIL) from both visualization and quantitative perspectives. Among them, ProIL imposes imitation on the RoIAlign-pooled $r \times r$ RoI representation as a whole, without modeling fine-grained correspondences. PIL establishes a PIL pathway within each pair, guiding each pixel to autonomously select imitated RoIAlign bins and dynamically adjust its learning weight.

We first visualize the gradients within each proposal induced by the imitation loss to analyze how supervision is distributed spatially. As shown in Fig. 16(b) and (c), for proposal-level imitation (CFINet-style), gradients exhibit a sparse binwise pattern consistent with the RoIAlign's sampling, reflecting that supervision is applied on the RoIAlign-pooled $r \times r$ bins rather than dense per-pixel supervision. In contrast, PIL produces dense, pixelwise varying gradients over the proposal region, evidencing imitation supervision at per-pixel granularity.

We further visualize how imitation gradients propagate back to the FPN feature maps by aggregating gradients from P2 and P3. As shown in Fig. 16(d) and (e), for PIL, the backpropagated gradients are concentrated on the proposal regions and align with the proposals' orientation, shape, and scale, indicating that our geometry and semantic-aligned imitation learning preserve the instance's geometric structure. By comparison, ProIL exhibits geometrically misaligned gradient responses, which may be attributed to the introduction of geometric bias from shape-mismatched exemplars. This also suggests that the detection RoI features not be rotated or transformed by the geometric normalization and semantic alignment.

Finally, we conduct experiments on both DOTA and SODA-A. For fair comparison, all experiments adopt Oriented R-CNN as the base detector with an identical feature bank construction (our REM-Bank), and differ only in the imitation granularity. As shown in Table IX, PIL consistently improves both mAP50 and mAP75 on both datasets, with a larger gain at mAP75 (+0.82 on DOTA and +2.2 on SODA-A), indicating that PIL facilitates geometry-aligned feature transfer, improving localization accuracy. ProIL is less stable: it increases mAP50 (+0.42) but reduces mAP75 (−0.42) on DOTA, while improving both metrics on SODA-A yet remains below PIL. This suggests that coarse global RoI-level imitation may introduce geometric ambiguity, as it tends to homogenize spatially distinct geometric cues.

TABLE IX
ABLATION ON IMITATION GRANULARITY. PROIL (CFINET-STYLE) VS PIL ON DOTA TEST SET AND SODA-A TEST SET

Methods	DOTA		SODA-A	
	mAP@0.5	mAP@0.75	mAP@0.5	mAP@0.75
Oriented R-CNN	75.87	50.32	70.7	28.6
ProIL	76.29 (+0.42)	49.9 (-0.42)	72.1 (+1.4)	30.0 (+1.4)
PIL (Ours)	**76.77** (+0.9)	**51.14** (+0.82)	**72.6** (+1.9)	**30.8** (+2.2)

TABLE X
GENERALIZATION OF PIL TO TRANSFORMER-STYLE REMOTE-SENSING DETECTORS ON DOTA TEST SET

Transformer-style baseline	ReDiffDet	RVSA	STD
mAP (Baseline)	73.88	78.96	79.68
mAP (+PIL)	74.90 (+1.02)	79.82 (+0.86)	79.81 (+0.13)

TABLE XI
ABLATION STUDY OF THE SUBMODULES IN CM^2-FPN

PIL	IC	IR	mAP
✓	×	×	76.77
✓	✓	×	77.41
✓	×	✓	77.39
✓	✓	✓	**78.36**

TABLE XII
ABLATION STUDY OF THE IC AGGREGATION

Components	Different Variants			
G		✓		
GI			✓	✓
LI				✓
mAP	75.89	75.94	77.07	**77.33**

TABLE XIII
ABLATION STUDY OF THE INTERLAYER SEMANTICS-SENSITIVE CHANNEL RECONSTRUCTION (IR)

Components	Different Variants			
GV		✓		✓
LW			✓	✓
mAP	75.89	76.79	76.78	**77.17**

g) Evaluation on transformer-style baselines: To examine whether the proposed PIL mechanism remains effective beyond the CNN-based baseline, we further evaluate PIL on several representative remote-sensing detectors that incorporate Transformer-style self-attention. Specifically, we conduct experiments on ReDiffDet [63], STD [71], and RVSA [74] using their publicly available codebases. For each baseline, we follow the released implementation and default configuration,

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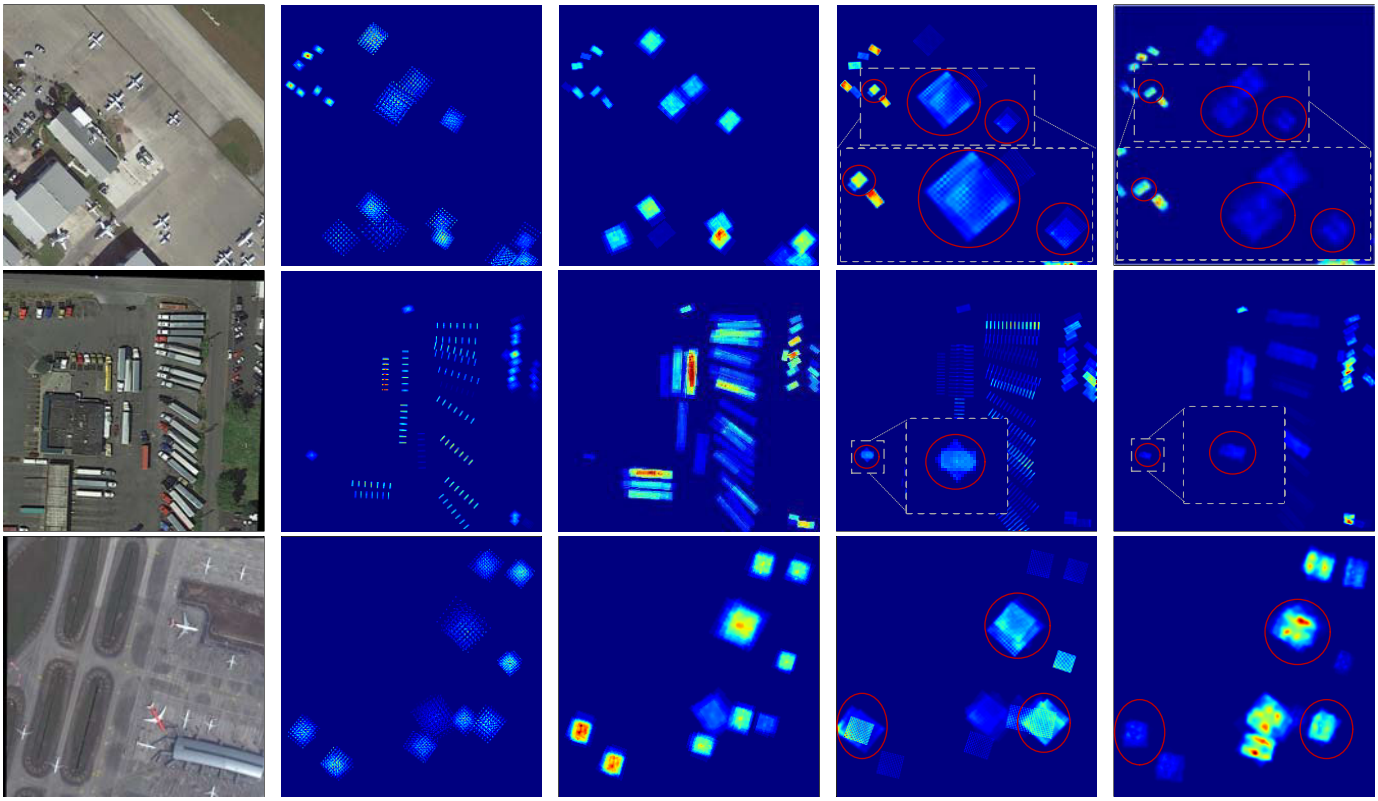


Fig. 16. Visualization of imitation gradients at different granularities. (a) Input image patch; (b) gradients within proposals induced by proposal-level imitation (ProIL); (c) gradients within proposals induced by PIL; (d) gradients on FPN features (P2+P3) induced by ProIL; (e) gradients on FPN features (P2+P3) induced by PIL.

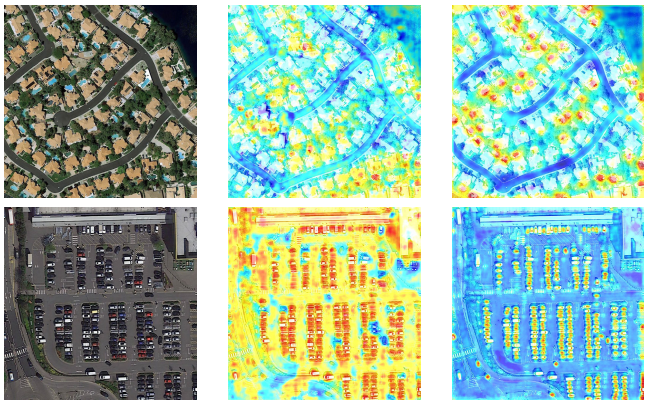


Fig. 17. Visualization of features enhanced by the proposed IC. (a) Input images. (b) Features without IC. (c) Features refined by IC.

and integrate PIL into the corresponding training pipeline. For ReDiffDet, we use the baseline performance reported in its public code repository. For STD, due to GPU memory constraints, we adopt a total mini-batch size of four for both the baseline and the PIL-augmented variant, using one image per GPU on four GPUs.

As shown in Table X, PIL improves all three baselines on the DOTA test set, increasing mAP by 1.02, 0.86, and 0.13 points, respectively. These demonstrate that exemplar-guided

TABLE XIV
COMPARISON RESULTS OF MULTISCALE OBJECT DETECTION ON DOTA VALIDATION SET

Methods	Backbone	mAP	AP_S	AP_M	AP_L
Baseline	R-50-FPN	74.32	56.66	72.36	78.65
PIL	R-50-FPN	74.84 (+0.52)	59.10 (+2.44)	72.79 (+0.43)	77.56 (-1.09)
IR	R-50-FPN	**75.41** (+1.09)	**60.60** (+3.94)	**72.85** (+0.49)	**78.83** (+0.18)

phenotype-aware imitation remains beneficial even with global self-attention-based feature modeling. However, the gain on STD is modest. This may be because it is specifically designed for ViT-based oriented detection: it explicitly decouples spatial transform factors and effectively harnesses the spatial transform potential of ViTs. As a result, it already offers strong rotation-sensitive feature modeling, and the incremental benefit brought by PIL becomes more marginal on this baseline. Overall, these results support that the proposed PIL mechanism remains beneficial on Transformer-style detectors, strengthening the generalizability of our motivation beyond the CNN-based baseline.

3) Evaluation of CM^2-FPN: The goal of CM^2-FPN is to progressively enhance intralayer semantic reliability and enrich high-level representations with the refined details from lower layers, thereby mitigating the loss of geometric details and enabling more effective cross-scale feature imitation. This

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is achieved through two key components: IC aggregation and interlayer semantics-sensitive channel reconstruction (IR).

a) Ablation study of submodules in CM²-FPN: To evaluate the submodule contributions, we conduct ablation experiments on top of the PIL mechanism. As shown in Table XI, incorporating IC with PIL mechanism increases the mAP to 77.41%, indicating its effectiveness in enhancing category discrimination of foreground and alleviating semantic confusion. Introducing IR raises the mAP to 77.39%, verifying that it is necessary for cross-scale imitation learning to enrich high-level representations with lower-layer geometric details. Significantly, their combination achieves the best performance of 78.36% mAP, highlighting their synergistic strengths in complementing the semantic-aware details in cross-scale imitation learning.

b) Evaluation of IC aggregation: To validate the effectiveness of IC in aggregating semantically structured global context through a GI module and purifying the semantic representation through an LI module, we conduct a series of ablation experiments by progressively introducing the original Gcblock (G) [77], GI, and LI. As shown in Table XII, GI improves the mAP to 77.07%, exceeding the G by 1.13%. This gain confirms the critical guidance role in aggregating semantic-aware context representation through embedding a compact structured semantic template via auxiliary segmentation. Building upon this, LI further improves detection performance, achieving the highest mAP of 77.33%, which indicates LI's ability to improve category discrimination through the multireception-field interaction between detection features and segmentation features. Overall, IC establishes a more reliable feature basis for subsequent IR.

Furthermore, Fig. 17 indicates that complex backgrounds can obscure instance features in the original FPN, whereas our IC enhances foreground discrimination and alleviates this interference.

c) Evaluation of interlayer semantics-sensitive channel reconstruction (IR): The goal of IR is to propagate and complement geometric details in a bottom-up way, through modeling channelwise dependencies between adjacent layers at both global and local spatial levels as interlayer context compensation vectors and reconstruction weights (GV and LW).

To verify the efficiency of these two submodules, we conduct ablation experiments. As shown in Table XIII, incorporating GV or GW individually yields mAPs of 76.79% and 76.78%, respectively. Significantly, their synergistic combination achieves 76.86% mAP, demonstrating IR's ability in compensating for geometric details in high-level features.

For rigorous verification, we categorize objects into small, medium, and large size, and evaluate detection performance on the DOTA validation set. Given that cross-scale detection inherently depends on different layers, this analysis directly reflects the module's capability to complement geometric detail in interlayer imitation. As shown in Table XIV, PIL mechanism yields a notable improvement on small-scale targets (+2.44 mAP) but a decline on large-scale targets (−1.09 mAP). This disparity stems from interlayer abstraction discrepancies: cross-scale imitation enhances the feature discriminability of weak small-scale objects, however, introduces ambiguous noise for already discriminative large-scale objects.

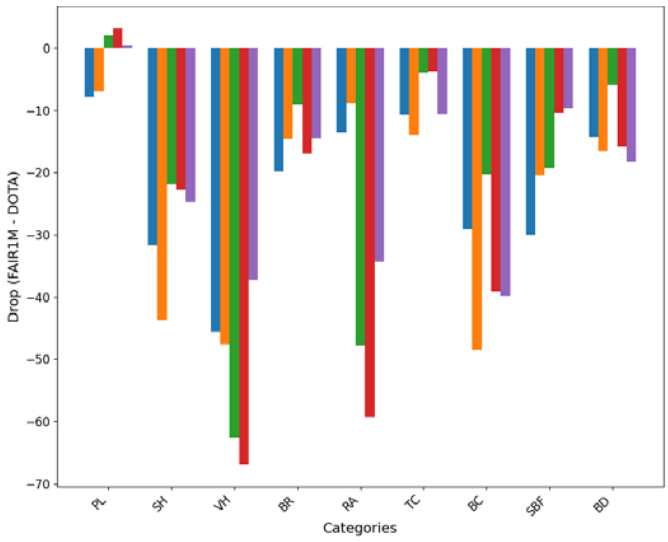


Fig. 18. Per-category AP drop under zero-shot cross-dataset transfer from DOTA to FAIR1M v2.0.

In contrast, IR outperforms the baseline in large-object mAP while further improving PIL mechanism's performance of small-object detection. This indicates that IR effectively mitigates the detail loss in cross-scale imitation mechanism.

G. Cross-Dataset Generalization in the Zero-Shot Setting

We evaluate cross-dataset generalization under domain shift through zero-shot cross-dataset transfer experiments. Specifically, we compare PIL-Net with representative oriented baselines, including Oriented R-CNN [36] (two-stage), S²A-Net [22] (one-stage), and transformer-based detectors (STD [71] and RVSA [74]). For each method, we use the official implementation and its publicly released DOTA-trained checkpoint, and directly evaluate it on unseen target datasets (HRSC2016 [75] and FAIR1M v2.0 [76]) without any fine-tuning.

For HRSC, which contains only the ship category, we evaluate each detector using its ship category outputs. As shown in Table XV, PIL-Net achieves 68.28 APship, corresponding to a 19.88-point drop from DOTA, which is substantially smaller than Oriented R-CNN, S²A-Net, and RVSA, while STD shows the smallest drop. A likely factor is that STD uses MAE-based self-supervised pretraining, which may improve robustness under distribution shifts.

For FAIR1M v2.0, we perform label-space alignment to ensure a fair comparison on overlapping categories.

1) The nine FAIR1M ship subcategories are merged into ship, the eleven airplane subcategories are merged into plane, and the ten vehicle subcategories are merged into a single vehicle category.
2) DOTA small vehicle and large vehicle are mapped to vehicle, and the vehicle AP is approximated by averaging their per-class APs, consistently for all methods.

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TABLE XV

ZERO-SHOT TRANSFER FROM DOTA TO HRSC2016 TEST SET WITHOUT FINE-TUNING. BOLD INDICATES THE SMALLEST DROP, AND UNDERLINE INDICATES THE SECOND SMALLEST DROP

Methods	Oriented R-CNN	S^2A-Net	STD	RVSA	PIL-Net (our)
DOTA (AP_{ship})	88.20	87.25	88.43	88.45	88.16
HRSC (AP_{ship})	49.54 (-38.66)	51.10 (-36.15)	**85.10 (-3.33)**	36.90 (-51.55)	<u>68.28 (-19.88)</u>

TABLE XVI

ZERO-SHOT TRANSFER FROM DOTA TO FAIR1M V2.0 VALIDATION SET WITHOUT FINE-TUNING. SAME NOTATION AS TABLE XV

Methods	Oriented R-CNN	S^2A-Net	STD	RVSA	PIL-Net (our)
DOTA (mAP)	78.85	76.72	80.86	79.66	79.14
FAIR1M (mAP)	55.39 (-23.46)	51.37 (-25.35)	<u>58.39 (-22.47)</u>	53.94 (-25.72)	**58.13 (-21.01)**

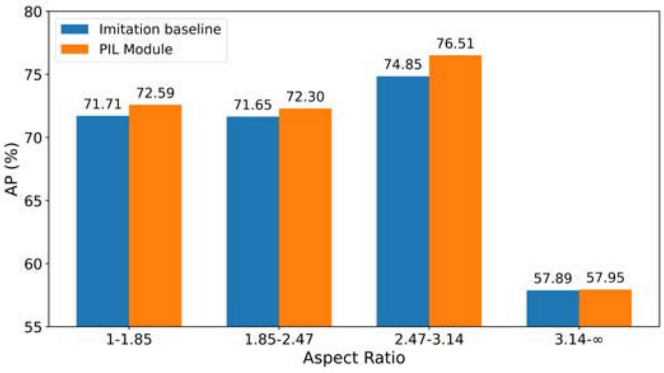


Fig. 19. Aspect-ratio-aware evaluation on DOTA validation set.

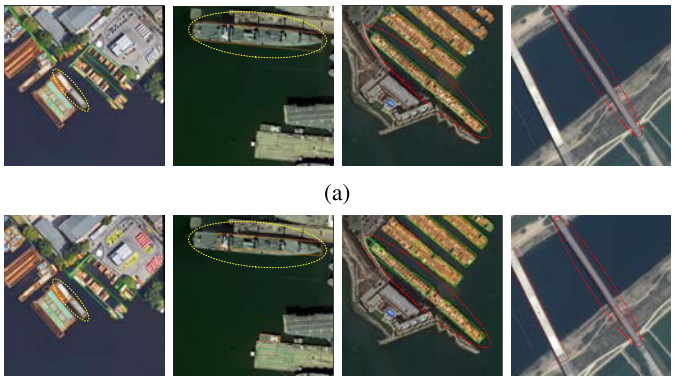


Fig. 20. Representative failure cases of PIL for instances with extreme aspect ratios. Boundary truncation is highlighted by yellow dashed boxes, and instance fragmentation is highlighted by red dashed boxes. (a) Ground truth annotations. (b) PIL failure cases.

3) Intersection is the only category without a DOTA counterpart and is excluded from evaluation. As shown in Table XVI, PIL-Net achieves 79.14 mAP on DOTA and 58.13 mAP on FAIR1M v2.0, corresponding to a drop of 21.01 points. This drop is the smallest among the compared approaches. In addition, we visualize the per-category performance drops in Fig. 18. Most categories degrade under domain shift, while PL remains stable and even shows slight gains for transformer-based methods as well as our PIL-Net. The largest attenuation generally occurs on VH and BC. Notably, PIL-Net yields the smallest drop on VH (−37.32 points) and shows the most stable performance on SBF (−9.71 points). These results suggest that PIL-Net has competitive zero-shot transferability under the evaluated cross-dataset shifts.

H. Failure Cases

While PIL consistently improves overall detection, we observe a slight AP drop on a few categories that frequently contain highly elongated instances (e.g., Bridge, Basketball Court, and Harbor), as reported in Table I. To quantify PIL's behavior across different object shapes, we further perform an aspect-ratio-aware evaluation on the DOTA validation set, and compare PIL with a global proposal-level imitation baseline (CFINet-style [18]). Specifically, we split ground truth instances into four groups according to quartiles of the ground truth aspect ratio distribution: $[1, 1.85)$,$[1.85, 2.47)$,$[2.47, 3.14)$,$[3.14, \infty)$, and report group-wise performance for both methods in Fig. 19. The results show that PIL yields consistent gains for low-to-moderate aspect ratio objects, whereas the improvement diminishes for the highest aspect ratio group $[3.14, \infty)$.

Furthermore, we provide representative failure cases in Fig. 20. For instances with extreme aspect ratios, the dominant errors manifested as boundary truncation and instance fragmentation. These failures may be due to the granularity mismatch in instance–exemplar imitation. Specifically, PIL performs PIL within each instance–exemplar pair, where each instance pixel adaptively selects the sub-bin from RoI-Aligned exemplar to imitate. However, for highly elongated instances, the fixed $r \times r$ RoI grid causes each teacher sub-bin to cover a large spatial region along the long axis, making the sub-bin feature a coarse spatial aggregation. Consequently, many distinct instance pixels are forced to imitate the same exemplar sub-bin, which weakens boundary-sensitive cues and hinders precise localization of object boundaries.

V. CONCLUSION

This work presented PIL-Net, designed to address geometric phenotype ambiguities in imitation learning-based arbitrary-oriented small-object detection. PIL-Net comprises

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a novel PIL mechanism and a complementary CM^2-FPN. The PIL mechanism explicitly disentangles geometric discrepancies and aligns semantic structures before imitation, thereby preserving diverse instance phenotypes and improving structure-consistent feature imitation. The CM^2-FPN further enhances cross-scale contextual interactions and mitigate geometric detail loss, enabling more effective cross-scale feature imitation. Extensive ablation and comparative experiments on challenging aerial image datasets (DOTA and DIOR-R) and a large-scale oriented small-object dataset (SODA-A) show that PIL-Net consistently achieves state-of-the-art performance, particularly for arbitrary-oriented small objects.

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